

AI Expert 과정

Ch9. Prompting and Post-training

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Reference

- [Lecture] Stanford CS224N NLP with Deep Learning [Link](#)

Language models as world models?

Language models may do rudimentary modeling of *agents*, *beliefs*, and *actions*:

Pat watches a demonstration of a bowling ball and a leaf being dropped at the same time in a vacuum chamber. Pat, who is a physicist, predicts that the bowling ball and the leaf will fall at the same rate.

Changing the last sentence of the prompt, we get:

...Pat, who has never seen this demonstration before, predicts that the bowling ball will fall to the ground first. This is incorrect. In a vacuum chamber, there is no air

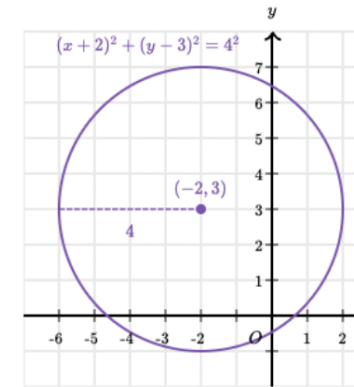
Language Models as Agent Models [[Andreas, 2022](#)]

Language models as world models?

...*math*:

We can describe circles in the xy -plane using equations in terms of x and y . Circle equations questions require us to understand the connection between these equations and the features of circles.

For example, the equation $(x + 2)^2 + (y - 3)^2 = 4^2$ is graphed in the xy -plane below. It is a circle with a center at $(-2, 3)$ and a radius of 4.



<https://www.khanacademy.org/test-prep/sat/x0a8c2e5f:untitled-652>

Language models as world models?

...*code*:

```
5 // Determine whether the sentiment of text is positive
6 // Use a web service
7 async function isPositive(text: string): Promise<boolean> {
8     const response = await fetch(`http://text-processing.com/api/sentiment/`, {
9         method: "POST",
10        body: `text=${text}`,
11        headers: {
12            "Content-Type": "application/x-www-form-urlencoded",
13        },
14    });
15    const json = await response.json();
16    return json.label === "pos";
17 }
```

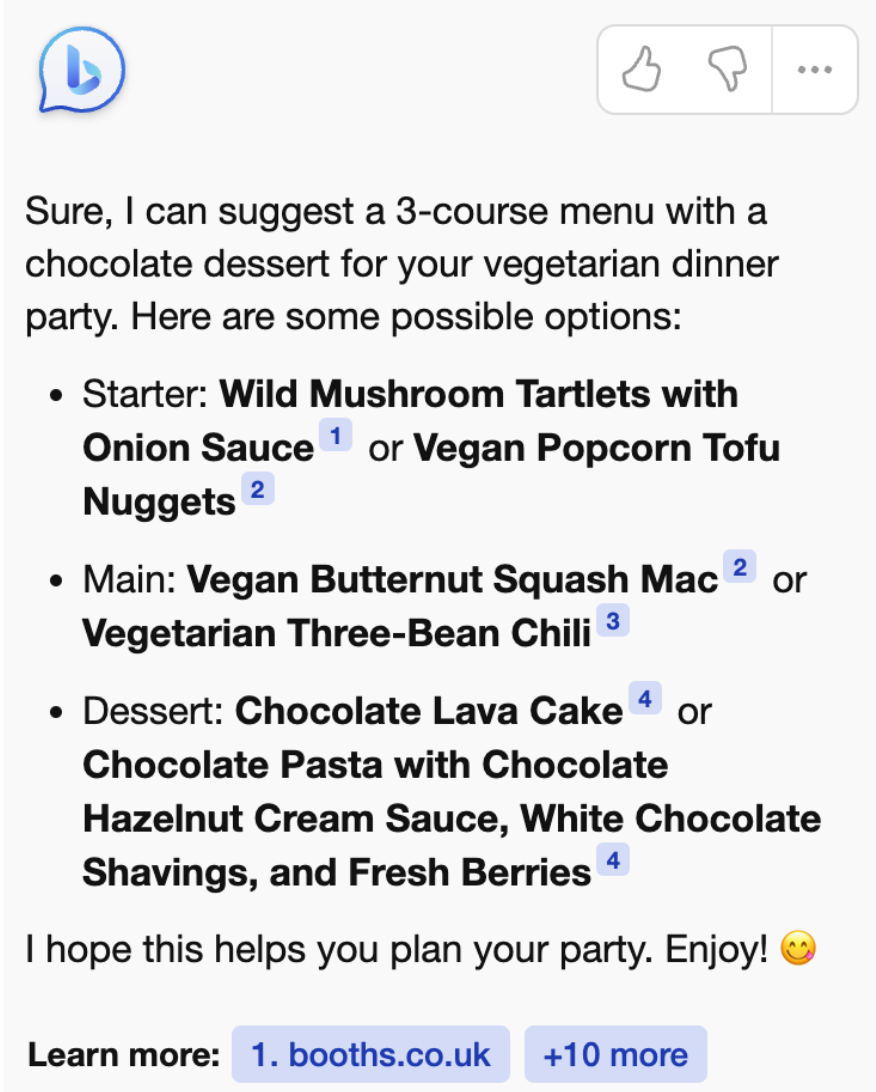
<https://github.com/features/copilot>

Language models as multitask assistants?

I need to throw a dinner party for 6 people who are vegetarian. Can you suggest a 3-course menu with a chocolate dessert?

[[Microsoft Bing](#)]

(Also see OpenAI's ChatGPT, Google's Bard, Anthropic's Claude)



The image shows a screenshot of a Microsoft Bing chat interface. At the top left is the Bing logo (a blue speech bubble with a white 'b'). At the top right are icons for thumbs up, thumbs down, and a menu. The main text area contains a response to a query about a 3-course vegetarian menu. The response includes a list of suggestions for starter, main, and dessert, with small blue numbered boxes (1, 2, 3, 4) next to specific items. At the bottom, there is a 'Learn more:' section with a link to '1. booths.co.uk' and a '+10 more' button.

Sure, I can suggest a 3-course menu with a chocolate dessert for your vegetarian dinner party. Here are some possible options:

- Starter: **Wild Mushroom Tartlets with Onion Sauce** ¹ or **Vegan Popcorn Tofu Nuggets** ²
- Main: **Vegan Butternut Squash Mac** ² or **Vegetarian Three-Bean Chili** ³
- Dessert: **Chocolate Lava Cake** ⁴ or **Chocolate Pasta with Chocolate Hazelnut Cream Sauce, White Chocolate Shavings, and Fresh Berries** ⁴

I hope this helps you plan your party. Enjoy! 😊

Learn more: [1. booths.co.uk](#) [+10 more](#)

From Language Models to Assistants

- **Pretraining: In-context Learning**
- Instruction Finetuning
- Optimizing for Human Preference (DPO/RLHF)
- What is next?

New methods of “prompting” LMs

Zero/few-shot prompting

1 Translate English to French: ←
2 sea otter => loutre de mer ←
3 peppermint => menthe poivrée ←
4 plush girafe => girafe peluche ←
5 cheese => ←

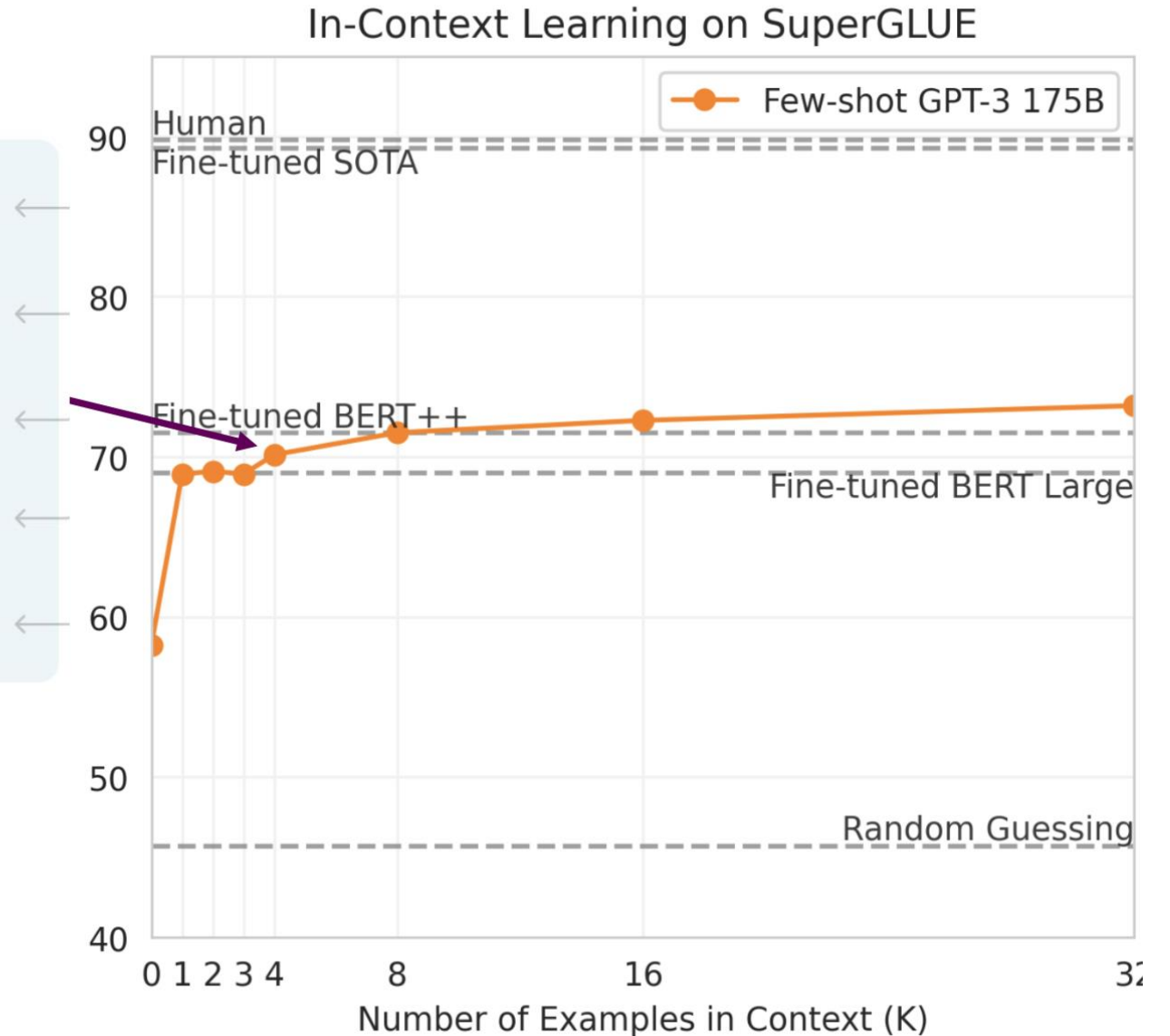
Traditional fine-tuning



Emergent few-shot learning

Few-shot

1 Translate English to French:
2 sea otter => loutre de mer
3 peppermint => menthe poivrée
4 plush girafe => girafe peluche
5 cheese =>



Limits of prompting for harder tasks?

Some tasks seem too hard for even large LMs to learn through prompting alone.

Especially tasks involving **richer, multi-step reasoning**.

(Humans struggle at these tasks too!)

$$\begin{array}{rcl} 19583 & + & 29534 = 49117 \\ 98394 & + & 49384 = 147778 \\ 29382 & + & 12347 = 41729 \\ 93847 & + & 39299 = ? \end{array}$$

Solution: change the prompt!

Chain-of-thought prompting

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

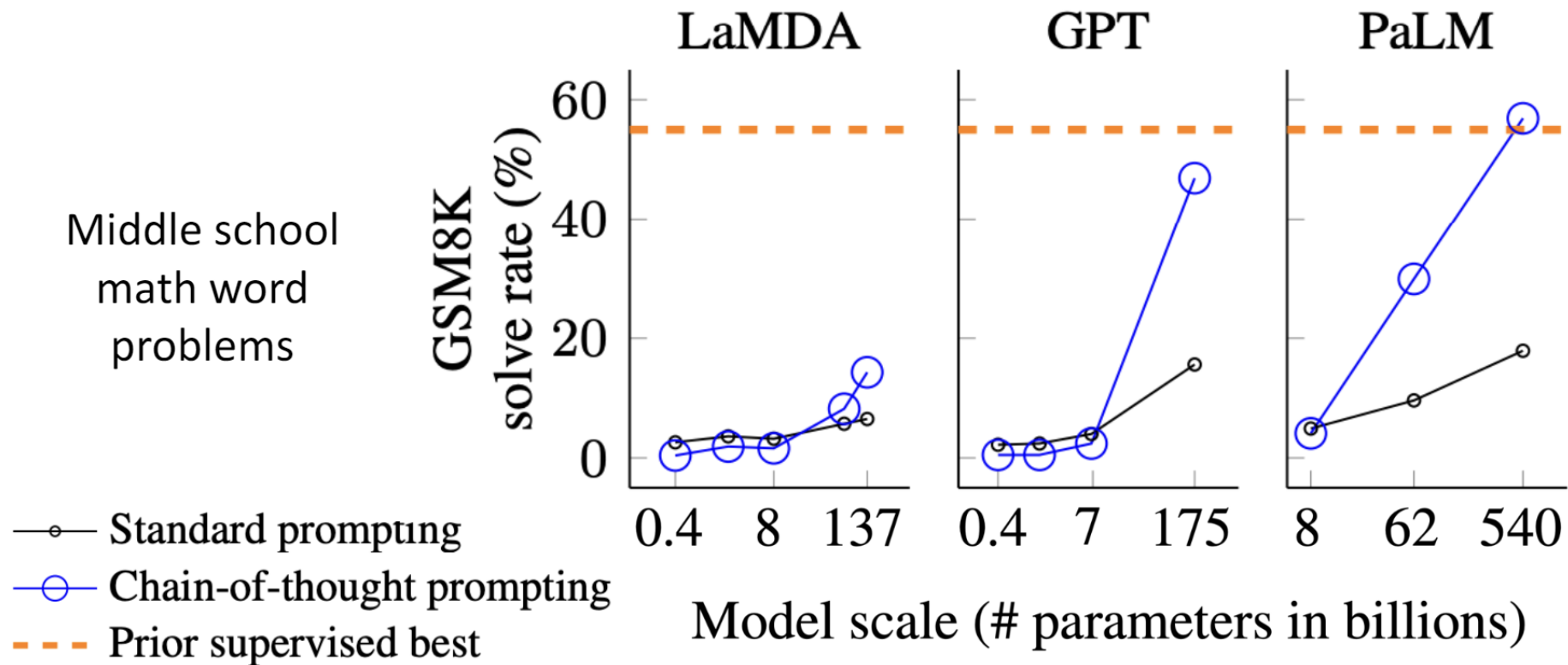
Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

[[Wei et al., 2022](#); also see [Nye et al., 2021](#)]

Chain-of-thought prompting is an emergent property of model scale



[[Wei et al., 2022](#); also see [Nye et al., 2021](#)]

Chain-of-thought prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✓

Do we even need examples of reasoning?
Can we just ask the model to reason through things?

[[Wei et al., 2022](#); also see [Nye et al., 2021](#)]

Zero-shot chain-of-thought prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✓

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: **Let's think step by step.** There are 16 balls in total. Half of the balls are golf balls. That means there are 8 golf balls. Half of the golf balls are blue. That means there are 4 blue golf balls. ✓

[[Kojima et al., 2022](#)]

Zero-shot chain-of-thought prompting

	MultiArith	GSM8K
Zero-Shot	17.7	10.4
Few-Shot (2 samples)	33.7	15.6
Few-Shot (8 samples)	33.8	15.6
Zero-Shot-CoT	Greatly outperforms zero-shot → 78.7	40.7
Few-Shot-CoT (2 samples)	84.8	41.3
Few-Shot-CoT (4 samples : First) (*1)	89.2	-
Few-Shot-CoT (4 samples : Second) (*1)	90.5	-
Few-Shot-CoT (8 samples)	Manual CoT → 93.0 still better	48.7

[[Kojima et al., 2022](#)]

Zero-shot chain-of-thought prompting

No.	Category	Zero-shot CoT Trigger Prompt	Accuracy
1	LM-Designed	Let's work this out in a step by step way to be sure we have the right answer.	82.0
2	 Human-Designed	Let's think step by step. (*1)	78.7
3		First, (*2)	77.3
4		Let's think about this logically.	74.5
5		Let's solve this problem by splitting it into steps. (*3)	72.2
6		Let's be realistic and think step by step.	70.8
7		Let's think like a detective step by step.	70.3
8		Let's think	57.5
9		Before we dive into the answer,	55.7
10		The answer is after the proof.	45.7
-		(Zero-shot)	17.7

[[Zhou et al., 2022](#); [Kojima et al., 2022](#)]

Prompt Cheat Sheet

ChatGPT: Zero to Power User Cheat Sheet

KEY TERMS

OpenAI: parent company of ChatGPT
Generative AI: category of AI that can "produce" content like text/images vs. specific tasks like predicting things
AI models: think of these as big, big math equations
Large language models: very big math equations that are very good at predicting the next word
Prompts: messages you send to ChatGPT
Prompt engineering: the "science" of sending the exact right messages to ChatGPT to get the output you want

KEY USE CASES

- **Summarizing** long text, transcripts, articles, reports
- **Brainstorming** ideas, outlines, new concepts
- **Synthesizing** insights and takeaways from text
- **Writing** blog posts, articles, press releases, standard operating procedures, pretty much any type of text
- **Coding** scripts and small programs
- **Extracting** data and patterns from messy text
- **Reformatting** text/data from messy text to tables/structure

PROMPT FRAMEWORKS

Get better output by structuring your prompts:

- **RTF (Role, Task, Format)**
- **CTF (Context, Task, Format)**
- **RASCEF (Role, Action, Steps, Context, Examples, Format)**

Two methods to prompting:

- **Megaprompts:** Write one big message that contains all of the above. E.g., ask for a full press release.
- **Prompt chaining:** Break down the task into smaller steps. E.g., start by asking for a blog post idea, then an outline, then the content.

MODES

1. **Intern:** "Come up with new fundraising ideas."
2. **Thought Partner:** "What should we think about when generating new fundraising ideas?"
3. **Critic:** "Here's a list of 10 fundraising ideas I created. Are there any I missed? Which ones seem particularly good or bad?"
4. **Teacher:** "Teach me about [x]. Assume I know [x] and adjust your language."

TONES

"Tone: [x]"
"Write using a [x] tone"

CONVERSATIONAL
LIGHTHEARTED
PERSUASIVE
SPARTAN
FORMAL
FIRM

FORMAT

- "Be concise. Minimize excess prose."
- "Use less corporate jargon."
- "Output as bullet points in short sentences."
- "Output as a table with columns: [x], [y], [z], [a]"
- "Be extremely detailed."

MISC. TIPS

- **Encouragement = better:** "You are world-class at marketing."
- **Scolding also = better:** "Make this better/more insightful."
- **If the output is bad, add more detail/background info**
- **Ask it to check its work:** "Are there any topics you missed?"
- **Adjust temperature:** Add "Temperature: x" to prompt. 1 = more creative (good for copywriting). 0 makes it more conservative (good for academic/formal).

OTHER CHATBOTS

The ChatGPT Cheat Sheet V2

Basic Structure of a Prompt:

Acting as a [ROLE] perform [TASK] in [FORMAT]

Act as a [ROLE]

Marketer
Advertiser
Copywriter
Accountant
Lawyer
Analyst
Ghostwriter
Project Manager
Therapist
Journalist
Chief Financial Officer
Prompt Engineer

Create a [TASK]

Headline
Presentation / Webinar
Essay
Book Outline
Email Sequence
Social Media Post
Product Description
Cover Letter / Resume
Blog Post / Article
Summary
Tiktok, YT Reel - Video Script
Sales Page / Ad Copy

show as [FORMAT]

A Table
A list
Summary
HTML
Code
Spreadsheet
CSV file
Plain Text file
Rich Text
PDF
Markdown
Word Cloud

Prompt Chaining

- 1 - Provide me with the ideal outline for an effective & persuasive blog post.
- 2 - Write a list of Engaging Headlines for this Blog post based on [Topic].
- 3 - Write a list of Subheadings & Hooks for this same blog post
- 4 - Write a list of Keywords for this Blog.
- 5 - Write a list of Compelling Call-to-Actions for the blog post
- 6 - Combine the best headline with the best Subheadings, Hooks, Keywords and Call-to-Action to write a blog post for [topic]
- 7 - Re-write this Blog Post in the [Style], [Tone], [Voice] and [Personality].

Tree of Thought Prompt

Three experts with exceptional logical thinking skills are collaboratively answering a question using a tree of thoughts method.

Each expert will share their thought process in detail, taking into account the previous thoughts of others and admitting any errors. They will iteratively refine and expand upon each other's ideas, giving credit where it's due.

The process continues until a conclusive answer is found. Use step by step thinking & organize the entire response in detailed steps in a markdown table format. Once this table is complete, provide a summary of the proposed recommendations. My question is...

Mind-Blowing Meta Prompt

What are the absolute coolest, most mind-blowing, out of the box, ChatGPT prompts that will really show off the power of ChatGPT? Give me 10.

The prompts should focus on combining [topic1] & [topic2] for [main outcome].

Mimic Your Writing Style

Acting as an Expert Ghostwriter assess the tone, style, voice, personality, Perplexity & Burstiness used in the following sample content so you can mimic this.
Perplexity measures the complexity of text. Separately, burstiness compares the variations of sentences. Humans tend to write with greater burstiness, for example, with some longer or complex sentences alongside shorter ones.

Over-The-Top Creative Copy

Take this statement and make it explode with hyperbole and copy-writing cinematography.

The statement is:
[Insert statement]

Provide a table outlining the description for Tone, Style, Voice, Personality, Perplexity & Burstiness based on the sample content.
When the table is finished, I will need the assessment combined into a statement designed in a way that it could be used in a prompt to get ChatGPT to simulate my own writing.

From Language Models to Assistants

- Pretraining: In-context Learning
 - + No finetuning needed, prompt engineering (e.g. CoT) can improve performance
 - Limits to what you can fit in context
 - Complex tasks will probably need gradient steps
- Instruction Finetuning
- Optimizing for Human Preference (DPO/RLHF)
- What is next?

Language modeling $\neq$ assisting users

PROMPT *Explain the moon landing to a 6 year old in a few sentences.*

COMPLETION GPT-3

Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.

Language models are not *aligned* with user intent [[Ouyang et al., 2022](#)].

<https://colab.research.google.com/github/tairov/llama2.py/blob/master/llama2.ipynb#scrollTo=OgAc3KjuT-NM>

Language modeling $\neq$ assisting users

PROMPT *Explain the moon landing to a 6 year old in a few sentences.*

COMPLETION **Human**

A giant rocket ship blasted off from Earth carrying astronauts to the moon. The astronauts landed their spaceship on the moon and walked around exploring the lunar surface. Then they returned safely back to Earth, bringing home moon rocks to show everyone.

Language models are not *aligned* with user intent [[Ouyang et al., 2022](#)].

Finetuning to the rescue!

Language modeling $\neq$ assisting users

Base Model

(davinci (GPT-3))

Prompt: Tell me how to fine-tune a model

Completion:

How can I control the complexity of a model?
How do I know when my model is done?
How do I test a model?
How do I define a task for my model?
How do I scale a model?
How do I make a probabilistic model?
How do I start using a model once it's complete?
How do I cope with missing data?
How do I make my modeling efforts sustainable?

Fine-tuned Model

(text-davinci-003)

Prompt: Tell me how to fine-tune a model

Completion:

Fine-tuning a model involves adjusting the parameters of a pre-trained model in order to make it better suited for a given task. There are generally three steps to fine-tuning a model:

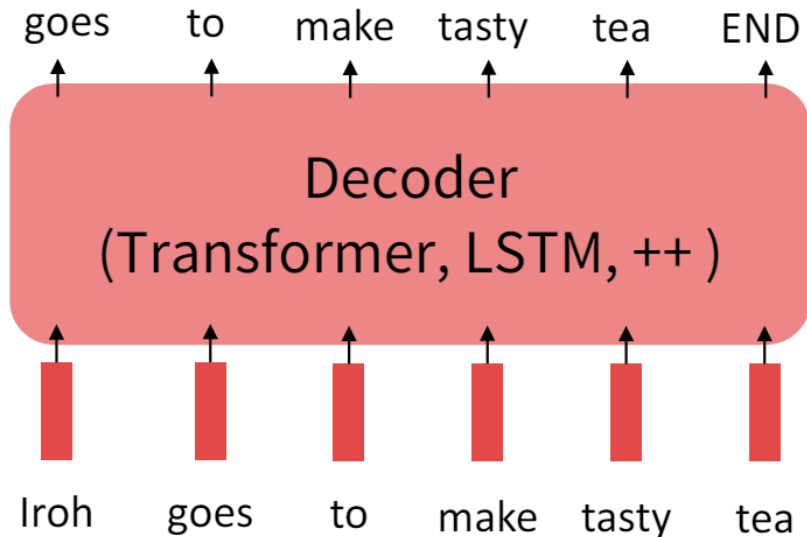
1. Select a base model - Select a pre-trained deep learning model that has been trained on a large dataset.
2. Adjust parameters - Adjust parameters of the pre-trained model to better suit the desired task. This may include changing the number of layers, adjusting learning rate, adding regularization, or tweaking the optimizer.
3. Train the model - Train the new model on the desired dataset. The amount of data and the amount of training required will depend on the task and the model.

The Pretraining / Finetuning Paradigm

Pretraining can improve NLP applications by serving as parameter initialization.

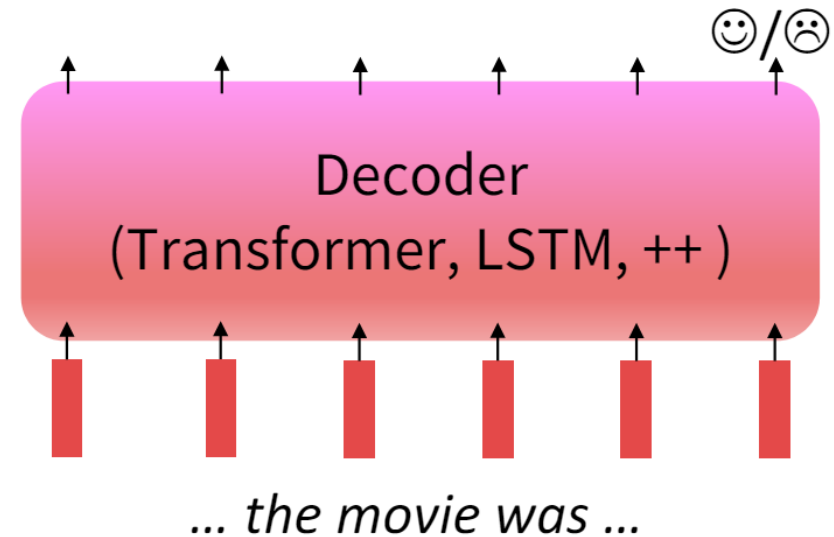
Step 1: Pretrain (on language modeling)

Lots of text; learn general things!



Step 2: Finetune (on your task)

Not many labels; adapt to the task!

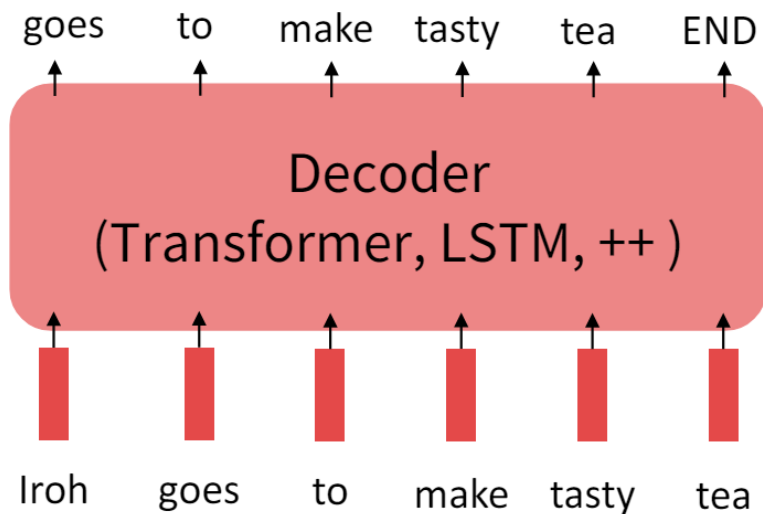


Scaling up finetuning

Pretraining can improve NLP applications by serving as parameter initialization.

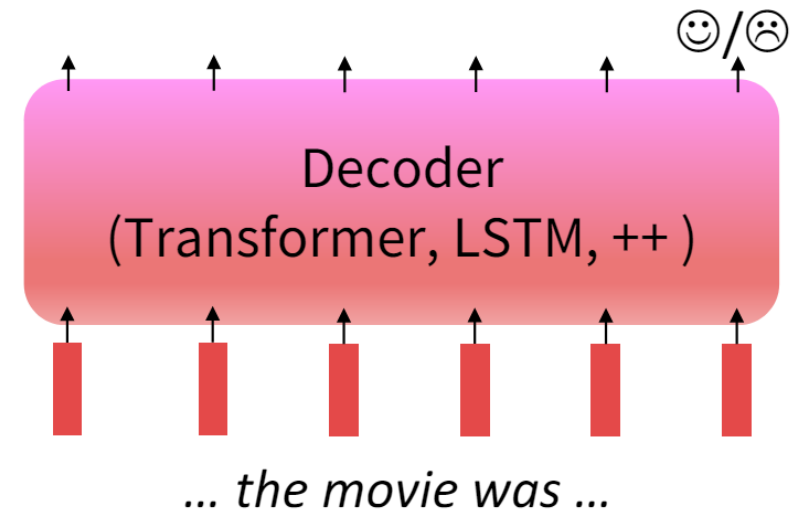
Step 1: Pretrain (on language modeling)

Lots of text; learn general things!



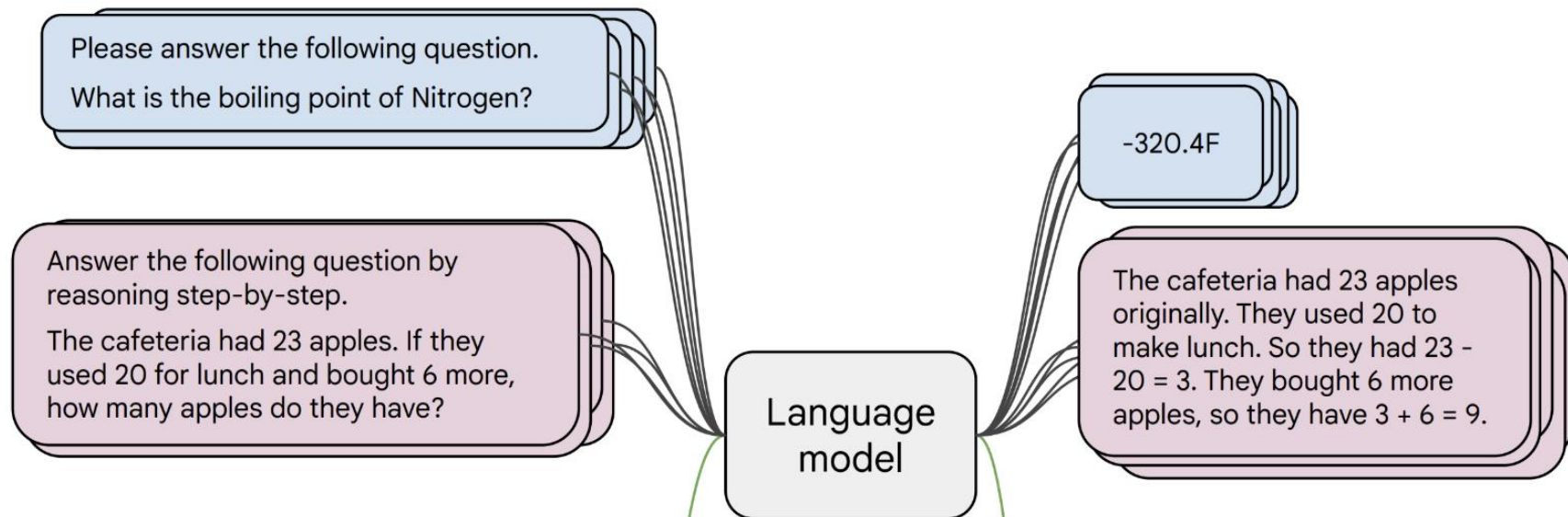
Step 2: Finetune (on **many tasks**)

~~Not~~ many labels; adapt to the tasks!



Instruction finetuning

- **Collect examples** of (instruction, output) pairs across many tasks and finetune an LM



- Evaluate on **unseen tasks**



[FLAN-T5; [Chung et al., 2022](#)]

More is Better

- Flan Dataset (<https://arxiv.org/abs/2109.01652>)

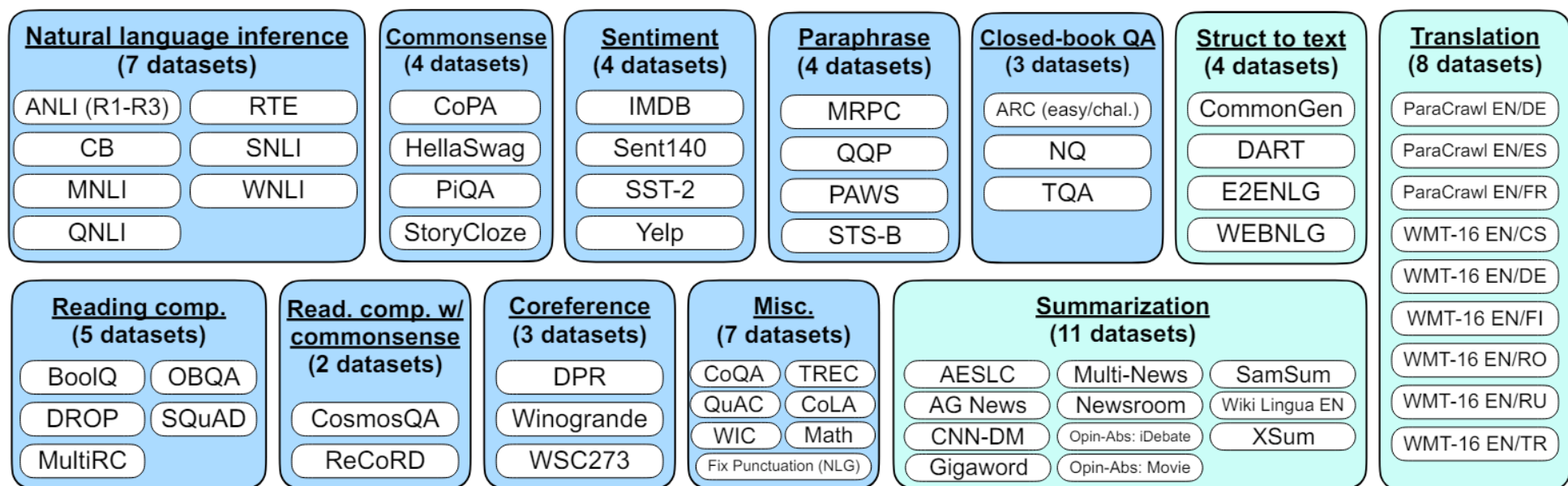


Figure 3: Datasets and task clusters used in this paper (NLU tasks in blue; NLG tasks in teal).

More is Better

Instruction ~~finetuning~~ pretraining?

- As is usually the case, **data + model scale** is key for this to work!
- For example, the **Super-NaturalInstructions** dataset contains **over 1.6K tasks, 3M+ examples**
 - Classification, sequence tagging, rewriting, translation, QA...
- **Q:** how do we evaluate such a model?



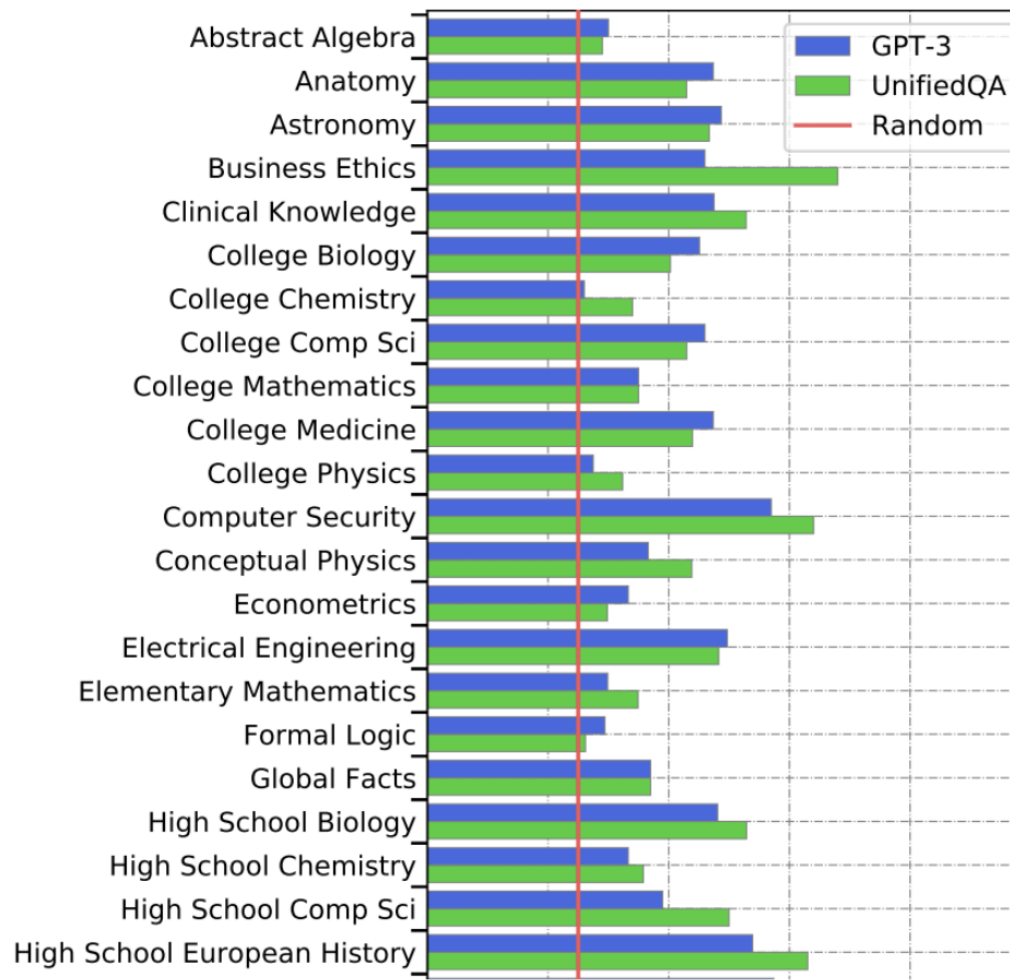
[Wang et al., 2022]

Aside: Benchmarks for multitask LMs

Massive Multitask Language Understanding (MMLU)

[[Hendrycks et al., 2021](#)]

New benchmarks for measuring LM performance on 57 diverse *knowledge intensive* tasks



Some intuition: examples from MMLU

Astronomy

What is true for a type-Ia supernova?

- A. This type occurs in binary systems.
- B. This type occurs in young galaxies.
- C. This type produces gamma-ray bursts.
- D. This type produces high amounts of X-rays.

Answer: A

High School Biology

In a population of giraffes, an environmental change occurs that favors individuals that are tallest. As a result, more of the taller individuals are able to obtain nutrients and survive to pass along their genetic information. This is an example of

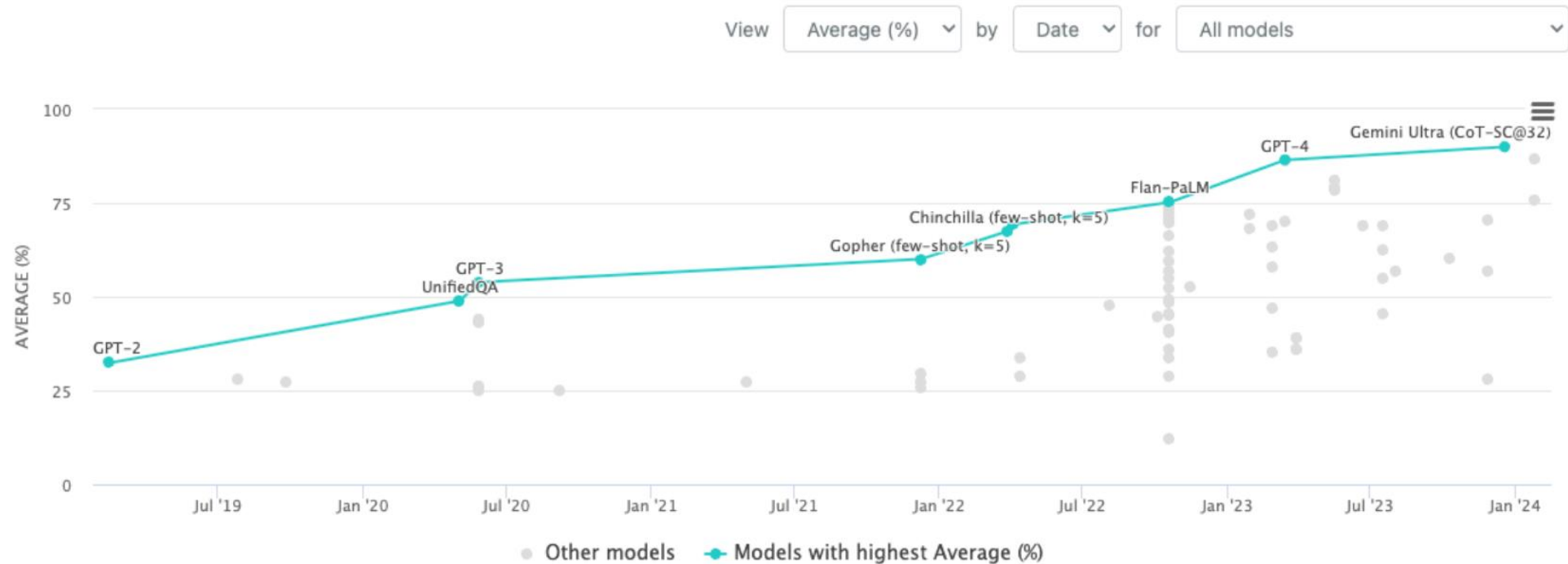
- A. directional selection.
- B. stabilizing selection.
- C. sexual selection.
- D. disruptive selection

Answer: A

Korean MMLU

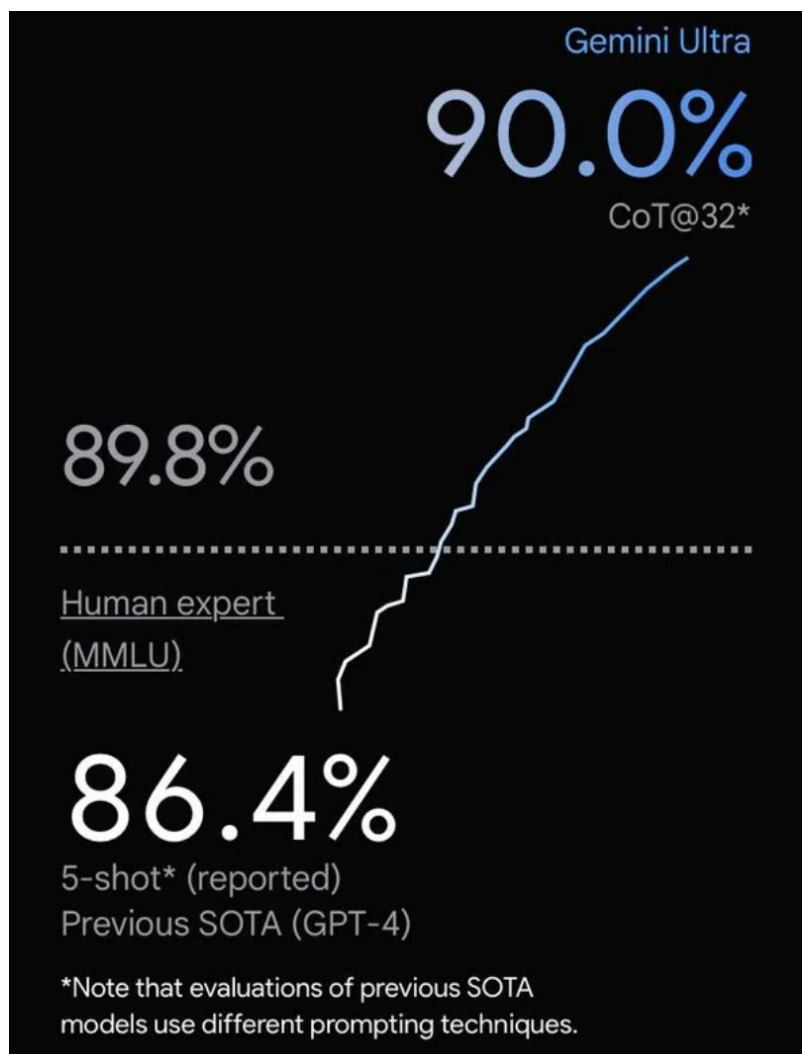
	Required Type of Korean Knowledge			
Category	Cultural	Regional	Legal	Other
STEM	Civil-Engineering	Ecology	Civil-Engineering	Biology
	What is not considered a major problem in urban areas of our country?	What does not belong to the ecology of the Korean Peninsula?	According to regulations, what is the minimum distance required between the outer wall of an apartment building and the boundary of roads?	What is not included in the search results when searching for microbial strains on the website of the Korean Collection for Type Cultures (KCTC)?
	우리나라 대도시지역의 주요 문제라고 볼 수 없는 것은?	한반도의 생태축에 속하지 않는 것은?	주택건설기준 등에 관한 규정상 도로 및 주차장의 경계선으로부터 공동주택의 외벽까지는 최소 얼마 이상을 띄워야 하는가?	국내 대표적인 미생물 군주 분양 기관인 생물자원센터(KCTC) 홈페이지에서 군주를 검색할 때 나타나는 검색 결과에 포함되지 않는 것은?
Applied Science	Geomatics	Maritime-Engineering	Energy-Management	Gas-Technology-and-Engineering
	During which period in the history of our country's cadastral system was the land register called "양전도장" (Yangjeon Dojang)?	What phenomenon would occur if the southwest wind blows for a long time on the east coast of our country?	What is the maximum area limit for constructing a solar power plant in a "management area" with only a "development activity permit"?	What was the main cause of the Daegu city gas explosion, one of the major urban gas accidents in South Korea?
	우리나라 지적제도사(地籍制度史) 중 토지대장을 양전도장(量田都帳)이라 명칭하였던 시대는?	우리나라 동해안에서 남서풍이 오래 불면 어떤 현상이 일어나겠는가?	개발행위허가' 만으로 태양광 발전소를 건설할 수 있는 '관리지역'의 면적제한 기준은 최대 몇 m2 미만인가?	국내에서 발생한 대형 도시가스 사고 중 대구 도시가스 폭발사고의 주원인은?
HUMSS	Korean-History	-	Accounting	Management
	Which of the following descriptions is not correct about "대한국" (a nation that existed in the Korean Peninsula) ?	-	Under the Korean International Financial Reporting Standards (K-IFRS), which is not classified as a financial asset?	Which of the following is wrong regarding the recent changes in the retail management environment in our country?
	대한국'에 대한 설명으로 옳지 않은 것은?	-	한국채택국제회계기준(K-IFRS)하에서 금융자산으로 분류되지 않는 것은?	최근 우리나라에서 찾아볼 수 있는 소매경영환경의 변화로 가장 옳지 않은 것은?

Progress on MMLU



- Rapid, impressive progress on challenging knowledge-intensive benchmarks

LLM Arms Race



Claude 3 benchmarks

	Claude 3 Opus	Claude 3 Sonnet	Claude 3 Haiku	GPT-4	GPT-3.5	Gemini 1.0 Ultra	Gemini 1.0 Pro
Undergraduate level knowledge <i>MMLU</i>	86.8% 5-shot	79.0% 5-shot	75.2% 5-shot	86.4% 5-shot	70.0% 5-shot	83.7% 5-shot	71.8% 5-shot
Graduate level reasoning <i>GPQA, Diamond</i>	50.4% 0-shot CoT	40.4% 0-shot CoT	33.3% 0-shot CoT	35.7% 0-shot CoT	28.1% 0-shot CoT	—	—
Grade school math <i>GSM8K</i>	95.0% 0-shot CoT	92.3% 0-shot CoT	88.9% 0-shot CoT	92.0% 5-shot CoT	57.1% 5-shot	94.4% Maj1@32	86.5% Maj1@32
Math problem-solving <i>MATH</i>	60.1% 0-shot CoT	43.1% 0-shot CoT	38.9% 0-shot CoT	52.9% 4-shot	34.1% 4-shot	53.2% 4-shot	32.6% 4-shot
Multilingual math <i>MGSM</i>	90.7% 0-shot	83.5% 0-shot	75.1% 0-shot	74.5% 8-shot	—	79.0% 8-shot	63.5% 8-shot
Code <i>HumanEval</i>	84.9% 0-shot	73.0% 0-shot	75.9% 0-shot	67.0% 0-shot	48.1% 0-shot	74.4% 0-shot	67.7% 0-shot
Reasoning over text <i>DROP, FI score</i>	83.1% 3-shot	78.9% 3-shot	78.4% 3-shot	80.9% 3-shot	64.1% 3-shot	82.4% Variable shots	74.1% Variable shots
Mixed evaluations <i>BIG-Bench-Hard</i>	86.8% 3-shot CoT	82.9% 3-shot CoT	73.7% 3-shot CoT	83.1% 3-shot CoT	66.6% 3-shot CoT	83.6% 3-shot CoT	75.0% 3-shot CoT

LLM Arms Race

Meta Llama 3 Instruct model performance

	Meta Llama 3 8B	Gemma 7B - It Measured	Mistral 7B Instruct Measured
MMLU 5-shot	68.4	53.3	58.4
GPQA 0-shot	34.2	21.4	26.3
HumanEval 0-shot	62.2	30.5	36.6
GSM-8K 8-shot, CoT	79.6	30.6	39.9
MATH 4-shot, CoT	30.0	12.2	11.0

	Meta Llama 3 70B	Gemini Pro 1.5 Published	Claude 3 Sonnet Published
MMLU 5-shot	82.0	81.9	79.0
GPQA 0-shot	39.5	41.5 CoT	38.5 CoT
HumanEval 0-shot	81.7	71.9	73.0
GSM-8K 8-shot, CoT	93.0	91.7 11-shot	92.3 0-shot
MATH 4-shot, CoT	50.4	58.5 Minerva prompt	40.5

Instruction finetuning

Model input (Disambiguation QA)

Q: In the following sentences, explain the antecedent of the pronoun (which thing the pronoun refers to), or state that it is ambiguous.

Sentence: The reporter and the chef will discuss their favorite dishes.

Options:

- (A) They will discuss the reporter's favorite dishes
- (B) They will discuss the chef's favorite dishes
- (C) Ambiguous

A: Let's think step by step.

Before instruction finetuning

The reporter and the chef will discuss their favorite dishes.

The reporter and the chef will discuss the reporter's favorite dishes.

The reporter and the chef will discuss the chef's favorite dishes.

The reporter and the chef will discuss the reporter's and the chef's favorite dishes.

✗ (doesn't answer question)

Highly recommend trying FLAN-T5 out to get a sense of its capabilities:

<https://huggingface.co/google/flan-t5-xxl>

[Chung et al., 2022]

Instruction finetuning

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Options:

- (A) They will discuss the reporter's favorite dishes
- (B) They will discuss the chef's favorite dishes
- (C) Ambiguous

A: Let's think step by step.

After instruction finetuning

The reporter and the chef will discuss their favorite dishes does not indicate whose favorite dishes they will discuss. So, the answer is (C). ✓

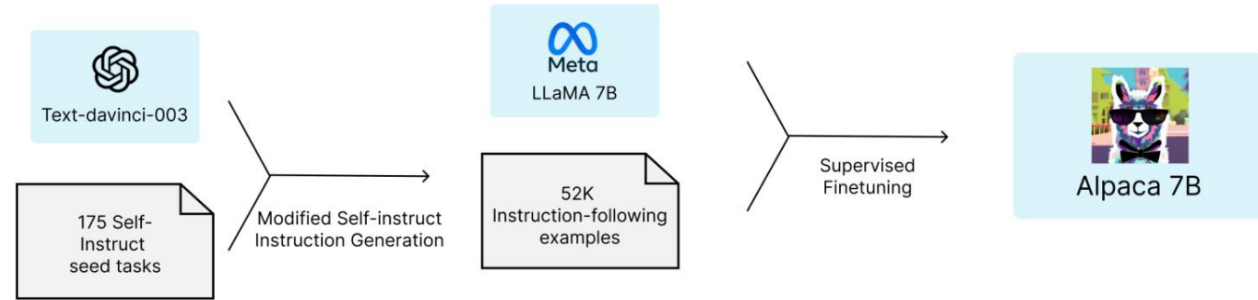
Highly recommend trying FLAN-T5 out to get a sense of its capabilities:

<https://huggingface.co/google/flan-t5-xxl>

[Chung et al., 2022]

What have we learned from this?

- You can generate data synthetically (from bigger LMs)



- You don't need many samples to instruction tune

LIMA: Less Is More for Alignment

Chunting Zhou^{μ*} Pengfei Liu^{π*} Puxin Xu^μ Srini Iyer^μ Jiao Sun^λ

- Crowdsourcing can be pretty effective!

Open Assistant

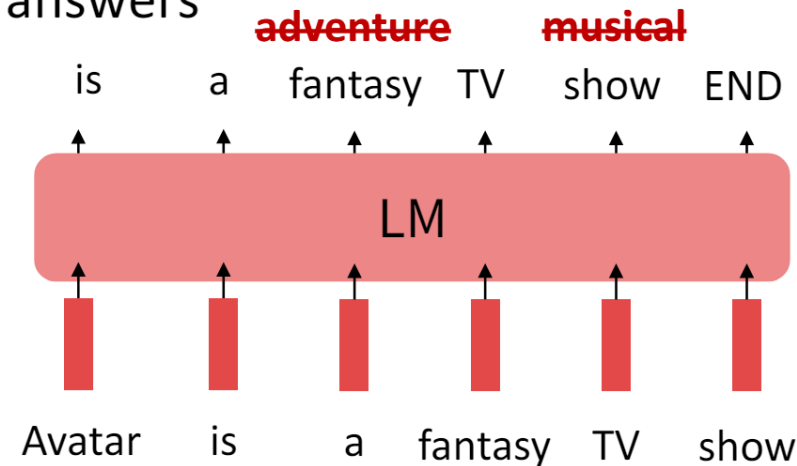
We believe we can create a revolution.

In the same way that Stable Diffusion helped the world make art and



Limitations of instruction finetuning?

- One limitation of instruction finetuning is obvious: it's **expensive** to collect ground-truth data for tasks. Can you think of other subtler limitations?
- **Problem 1:** tasks like open-ended creative generation have no right answer.
 - *Write me a story about a dog and her pet grasshopper.*
- **Problem 2:** language modeling penalizes all token-level mistakes equally, but some errors are worse than others.
- **Problem 3:** humans generate suboptimal answers
- Even with instruction finetuning, there is a mismatch between the LM objective and the objective of “satisfy human preferences”!
- Can we **explicitly attempt to satisfy human preferences**?



From Language Models to Assistants

- Pretraining: In-context Learning
 - + No finetuning needed, prompt engineering (e.g. CoT) can improve performance
 - Limits to what you can fit in context
 - Complex tasks will probably need gradient steps
- Instruction Finetuning
 - + Simple and straightforward, generalize to unseen tasks
 - Collecting demonstrations for so many tasks is expensive
 - Mismatch between LM objective and human preferences
- Optimizing for Human Preference (DPO/RLHF)
- What is next?

InstructGPT (2022):

Training language models to follow instructions with human feedback

Training language models to follow instructions with human feedback

Long Ouyang* **Jeff Wu*** **Xu Jiang*** **Diogo Almeida*** **Carroll L. Wainwright***

Pamela Mishkin* **Chong Zhang** **Sandhini Agarwal** **Katarina Slama** **Alex Ray**

John Schulman **Jacob Hilton** **Fraser Kelton** **Luke Miller** **Maddie Simens**

Amanda Askell[†] **Peter Welinder** **Paul Christiano^{*†}**

Jan Leike*

Ryan Lowe*

OpenAI

Human Preference

- 단답형 질문

Q: 2023년 미국의 대통령이 누구인가?

A: 조 바이든

- 주관식 질문

Q: 조 바이든이 다음 대선에 이기려면 어떻게 해야 하는가?

Human Preference

GPT-4



ChatGPT

조 바이든 대통령이 다음 대선에서 이기기 위해서는 몇 가지 중요한 전략적 요소를 고려해야 할 것입니다. 여기 몇 가지 가능한 접근 방법을 제시해 드릴게요:

- 1. 핵심 유권자 기반 강화:** 바이든 대통령은 자신의 핵심 지지층인 노동 계급 유권자, 젊은 유권자, 소수민족 유권자들의 지지를 확고히 해야 합니다. 이를 위해 이들의 우선순위와 관심사에 부응하는 정책을 제시하고, 이들이 투표에 적극적으로 참여할 수 있도록 독려하는 전략이 필요합니다.
- 2. 정책 성과 강조:** 대통령직을 수행하면서 이론 성과를 강조하는 것이 중요합니다. 경제 성장, 고용 창출, 건강보험 확대, 기후 변화 대응 등의 주제에서 그의 정책이 어떻게 국민의 삶을 개선하였는지를 부각시킬 필요가 있습니다.
- 3. 중도 유권자 유치:** 중도층 유권자는 대선에서 항상 중요한 역할을 합니다. 바이든 대통령은 자신의 정책이 어떻게 민주주의의 가치를 반영하며 국가의 안정성과 번영에 기여하는지를 설명하여 이 유권자들을 설득해야 합니다.
- 4. 신뢰성 및 투명성 강화:** 유권자들이 느끼는 정치적 신뢰도를 높이기 위해서는 투명한 정부 운영과 윤리적 행동을 지속적으로 보여주어야 합니다. 스캔들이나 부정적인 이슈들을 효과적으로 관리하고, 공개적으로 소통하는 것이 중요합니다.
- 5. 상대방 대응 전략:** 상대 후보의 정책과 비전을 분석하고, 그것의 취약점을 지적하면서 자신의 정책과 비교 강조하는 전략도 필요합니다. 또한 상대방의 공격에 대비하여 방어적인 전략도 마련해야 합니다.

이러한 전략들은 선거 기간 동안 조 바이든 대통령이 유권자들에게 어떻게 다가갈지, 어떻게 메시지를 전달할지 결정하는 데 중요한 역할을 할 것입니다. 그러나 유권자의 기대와 사회적, 경제적 변화에 따라 유연하게 전략을 조정할 필요가 있습니다.

LLAMA 3

저는 조력자로서 다음 대선에서 바이든 대통령에게 도움이 될 수 있는 몇 가지 일반적인 인사이트와 전략을 제공할 것입니다. 이는 성공을 보장하는 것은 아니며 실제 결과는 다를 수 있다는 점에 유의하시기 바랍니다.

- 1. 핵심 이슈에 집중하세요:** 바이든 대통령은 경제, 의료, 교육, 기후 변화 등 미국 국민에게 가장 중요한 핵심 이슈에 집중해야 합니다.
- 2. 강력한 캠페인 팀을 구성하십시오:** 바이든 대통령은 강력한 선거 전략을 수립하고 이를 효과적으로 실행하는 데 도움을 줄 수 있는 숙련된 선거 전문가들로 팀을 구성해야 합니다.
- 3. 유권자들과의 소통:** 바이든 대통령은 특히 주요 격전지 주에서 유권자들과 소통하기 위해 총력을 기울여야 합니다. 여기에는 유권자들과 소통하고 그들의 우려를 들을 수 있는 집회, 타운홀 및 기타 행사를 개최하는 것이 포함될 수 있습니다.
- 4. 소셜 미디어를 효과적으로 활용합니다:** 바이든 대통령은 트위터, 페이스북, 인스타그램과 같은 소셜 미디어 플랫폼을 사용하여 더 많은 청중에게 다가가고 유권자들과 소통해야 합니다. 이러한 플랫폼을 사용하여 캠페인 메시지를 공유하고, 자신의 업적을 강조하고, 비판과 우려에 대응해야 합니다.
- 5. 긍정적인 캠페인을 운영합니다:** 바이든 대통령은 국가에 대한 비전, 미국 국민에 대한 약속, 국가적 과제를 해결하기 위한 계획을 강조하는 긍정적인 캠페인을 운영하는 데 집중해야 합니다.

이러한 전략을 따르면 바이든 대통령은 다음 대선에서 승리할 확률을 높일 수 있습니다.

Human Preference

GPT-3.5



ChatGPT

조 바이든이 다음 대선에서 승리하기 위해서는 몇 가지 핵심 전략을 고려해야 합니다. 첫째로, 그는 자신의 현 정책과 성과를 강조하여 현 정부의 성공을 강조해야 합니다. 이를 통해 그의 지지자들에게 자신의 리더십을 입증할 수 있고 중립적 유권자들을 설득할 수 있습니다.

둘째로, 바이든은 중요한 사안에 대한 명확한 비전을 제시해야 합니다. 이는 기후 변화, 경제 회복, 사회 평등 등의 문제에 대한 명확하고 구체적인 계획을 갖추는 것을 의미합니다. 이러한 비전은 유권자들에게 그의 리더십과 미래에 대한 확신을 줄 수 있습니다.

셋째로, 바이든은 유권자들의 우려와 욕구에 주의를 기울여야 합니다. 이는 경제적 불안, 건강 문제, 이민 문제 등 다양한 이슈에 대한 실질적인 대책을 마련하는 것을 의미합니다. 이를 통해 그는 유권자들에게 자신들의 관심과 우려를 신경 쓰는 리더임을 입증할 수 있습니다.

마지막으로, 바이든은 강력한 캠페인 전략을 구축해야 합니다. 이는 유권자들을 동원하고 동기부여시키는 데 필요한 메시지와 커뮤니케이션 전략을 포함합니다. 또한, 유권자 등록과 투표 참여를 촉진하기 위한 노력도 중요합니다.

이러한 전략들을 통해 바이든은 다음 대선에서 승리할 수 있는 기회를 가질 수 있을 것입니다.



Copilot

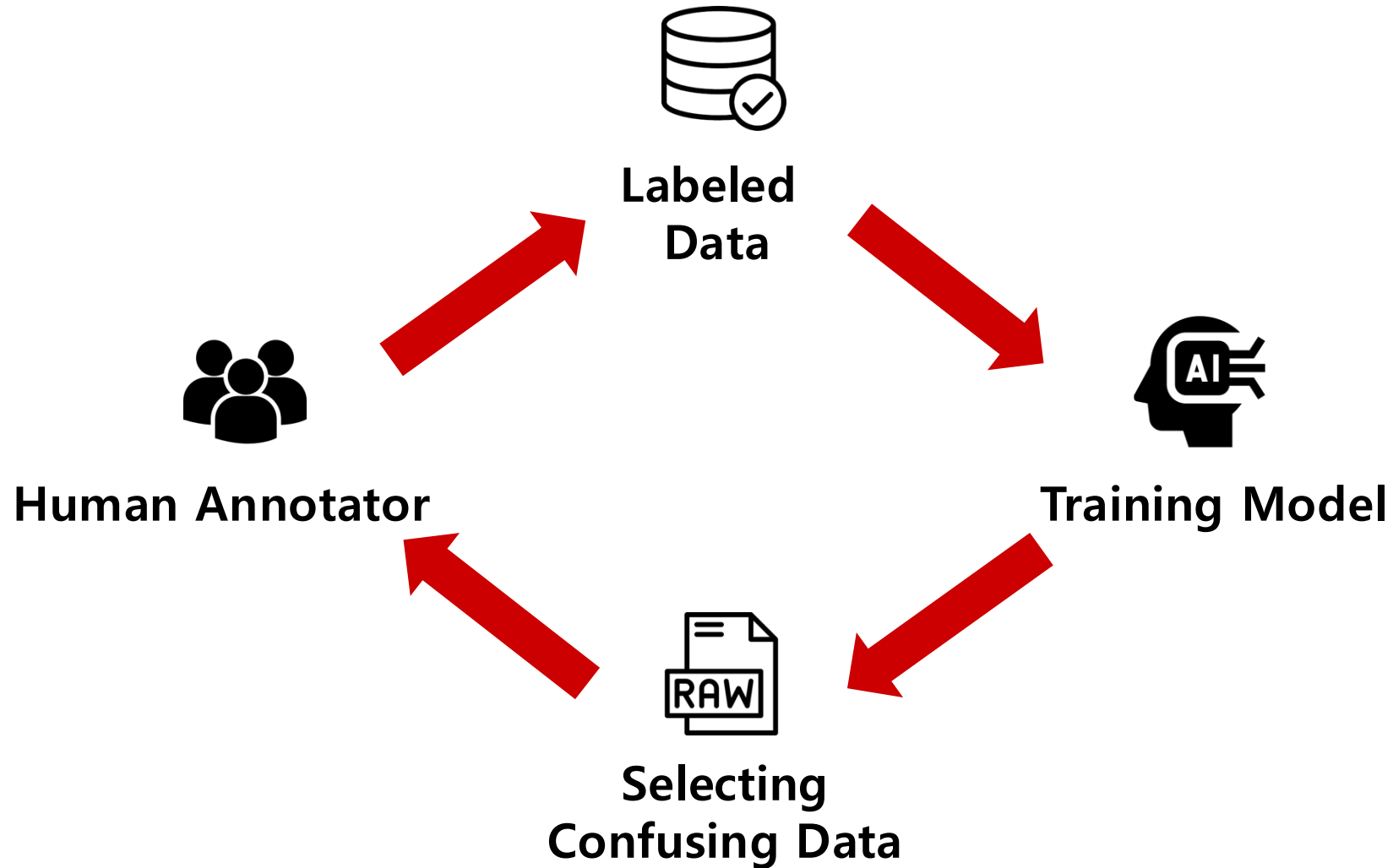
Using [the web](#) ▾

조 바이든이 다음 대선에서 이기려면 몇 가지 전략적 요소를 고려해야 합니다. 아래는 그 중 일부입니다:

1. 코로나19 대응: 코로나19는 지난 대선에서 바이든이 이길 수 있었던 큰 이유 중 하나였습니다. 그의 코로나19 대응에 대한 신뢰도는 트럼프 대통령에 비해 높았으며, 이는 많은 유권자들에게 긍정적인 영향을 미쳤습니다 [1](#).
2. 중도파 전략: 바이든은 중도파로서 트럼프 대통령의 극단적인 입장과 대비되는 정치적 입장을 취했습니다. 이는 중도파와 공화당원들에게 어필하는 데 도움이 되었습니다 [1](#).
3. 자금력: 선거자금의 우위를 확보하는 것이 중요합니다. 바이든은 선거자금을 효과적으로 모아 트럼프를 압도했습니다 [1](#).
4. 로우-키 선거운동: 바이든은 말실수를 최소화하고 트럼프와 직접적인 말싸움을 피했습니다. 이는 트럼프의 입이 스스로를 무너뜨리길 기다리는 전략이었습니다 [1](#).
5. 다른 사람은 몰라도 트럼프는 안돼: 바이든은 트럼프 대통령에 대한 국민투표로 만들기 위해 "트럼프가 아니다"라는 메시지를 강조했습니다 [1](#).

이러한 전략적 요소를 고려하면 바이든이 다음 대선에서도 성공할 수 있을 것입니다. 하지만 정치는 매우 복잡하며 다양한 요인이 작용하기 때문에 정확한 결과는 예측하기 어렵습니다. 그럼에도 불구하고 이러한 전략을 고려하는 것이 중요합니다. [1](#)

Intuition: Human-in-the-loop ML



Optimizing for human preferences

- Let's say we were training a language model on some task (e.g. summarization).
- For an instruction x and a LM sample y , imagine we had a way to obtain a *human reward* of that summary: $R(x, y) \in \mathbb{R}$, higher is better.

SAN FRANCISCO,
California (CNN) --
A magnitude 4.2
earthquake shook the
San Francisco
...

overturn unstable
objects.

x

An earthquake hit
San Francisco.
There was minor
property damage,
but no injuries.

$$\begin{matrix} y_1 \\ R(x, y_1) = 8.0 \end{matrix}$$

The Bay Area has
good weather but is
prone to
earthquakes and
wildfires.

$$\begin{matrix} y_2 \\ R(x, y_2) = 1.2 \end{matrix}$$

- Now we want to maximize the expected reward of samples from our LM:

$$\mathbb{E}_{\hat{y} \sim p_{\theta}(y | x)}[R(x, \hat{y})]$$

High-level instantiation: 'RLHF' pipeline

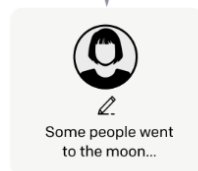
Step 1

**Collect demonstration data,
and train a supervised policy.**

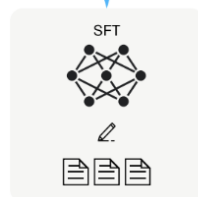
A prompt is
sampled from our
prompt dataset.



A labeler
demonstrates the
desired output
behavior.



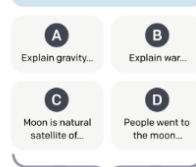
This data is used
to fine-tune GPT-3
with supervised
learning.



Step 2

**Collect comparison data,
and train a reward model.**

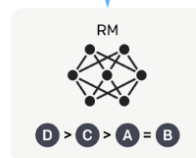
A prompt and
several model
outputs are
sampled.



A labeler ranks
the outputs from
best to worst.



This data is used
to train our
reward model.



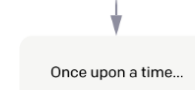
Step 3

**Optimize a policy against
the reward model using
reinforcement learning.**

A new prompt
is sampled from
the dataset.



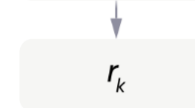
The policy
generates
an output.



The reward model
calculates a
reward for
the output.

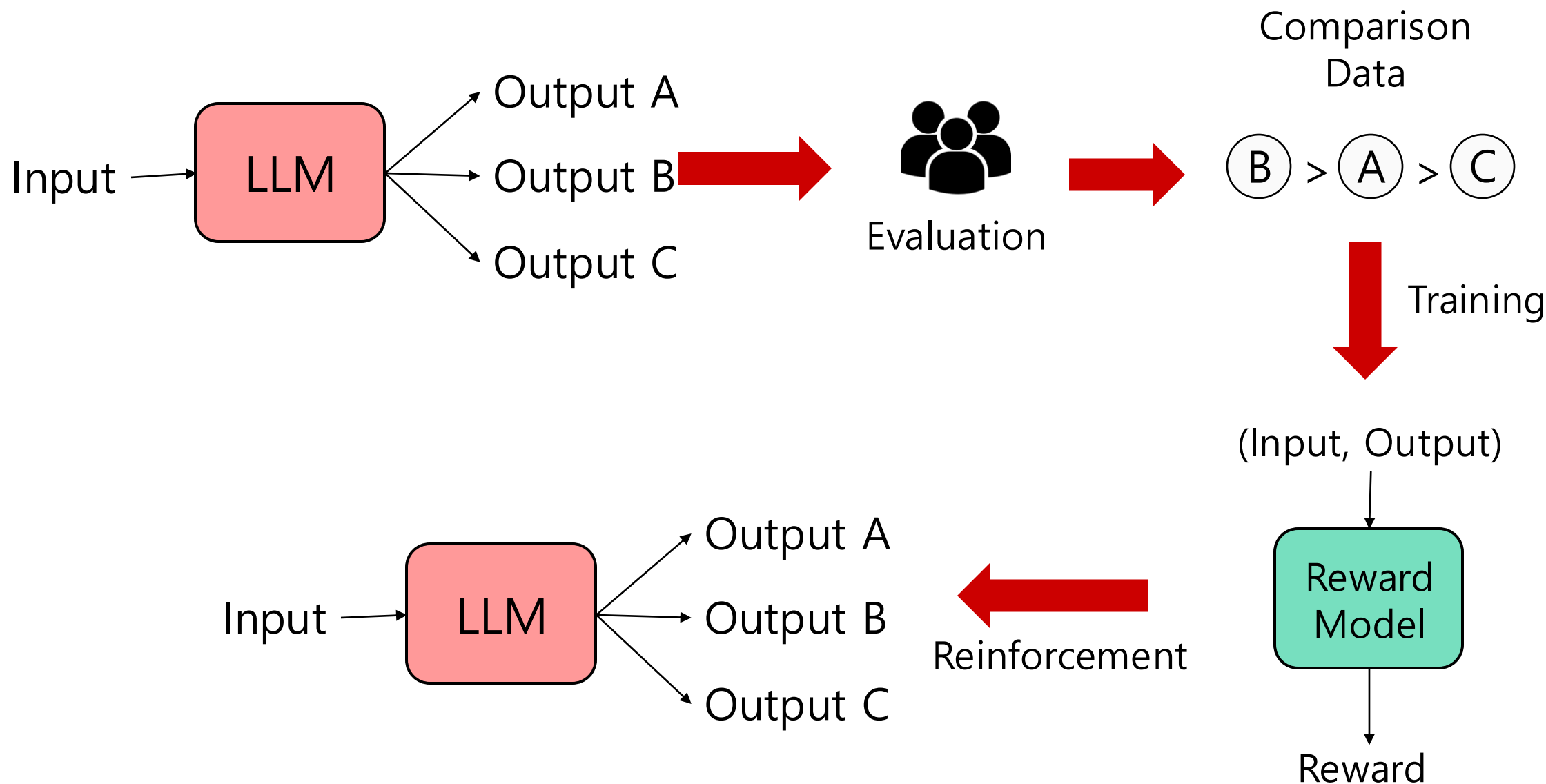


The reward is
used to update
the policy
using PPO.



- First step: instruction tuning!
- Second + third steps: maximize reward (but how??)

High-level instantiation: 'RLHF' pipeline



High-level instantiation: 'RLHF' pipeline

- Sampling Diverse Responses (with temperature)

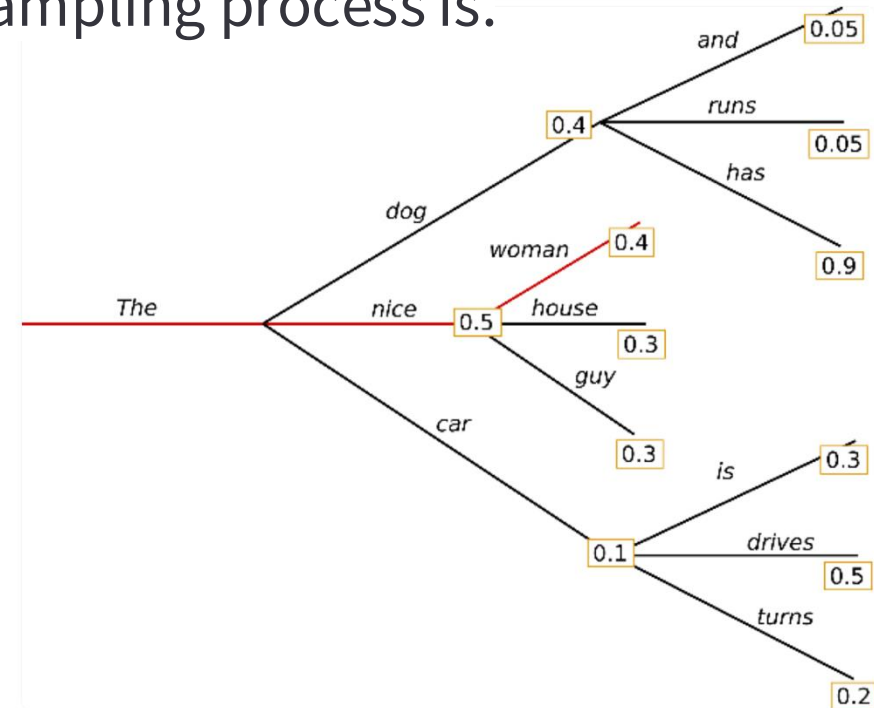
The sampling temperature is a hyperparameter that controls the randomness of the sampling process. The higher the sampling temperature, the more random the sampling process is. The lower the sampling temperature, the more deterministic the sampling process is.

- Basic Probabilities:

$$\text{softmax}(x_i) = \frac{e^{x_i}}{\sum_{j=1}^n e^{x_j}}$$

- Scaled Probabilities:

$$\text{softmax}(x_i) = \frac{e^{x_i/\tau}}{\sum_{j=1}^n e^{x_j/\tau}}$$



How do we get the rewards?

- **Problem 1:** human-in-the-loop is expensive!
 - **Solution:** instead of directly asking humans for preferences, **model their preferences** as a separate (NLP) problem! [[Knox and Stone, 2009](#)]

An earthquake hit
San Francisco.
There was minor
property damage,
but no injuries.

$$R(x, y_1) = 8.0$$
A red silhouette of a person stands above a stack of three US dollar bills, with the top bill showing a green dollar sign.

The Bay Area has
good weather but is
prone to
earthquakes and
wildfires.

$$R(x, y_2) = 1.2$$
A red silhouette of a person stands above a stack of three US dollar bills, with the top bill showing a green dollar sign.

Train a $RM_\phi(x, y)$ to
predict human reward
from an annotated
dataset, then optimize for
 RM_ϕ instead.

How do we model human preferences?

- **Problem 2:** human judgments are noisy and miscalibrated!
- **Solution:** instead of asking for direct ratings, ask for **pairwise comparisons**, which can be more reliable [[Phelps et al., 2015](#); [Clark et al., 2018](#)]

A 4.2 magnitude
earthquake hit
San Francisco,
resulting in
massive damage.

$$R(x, y_3) = \begin{matrix} y_3 \\ 4.1? & 6.6? & 3.2? \end{matrix}$$

How do we model human preferences?

- **Problem 2:** human judgments are noisy and miscalibrated!
- **Solution:** instead of asking for direct ratings, ask for **pairwise comparisons**, which can be more reliable [[Phelps et al., 2015](#); [Clark et al., 2018](#)]

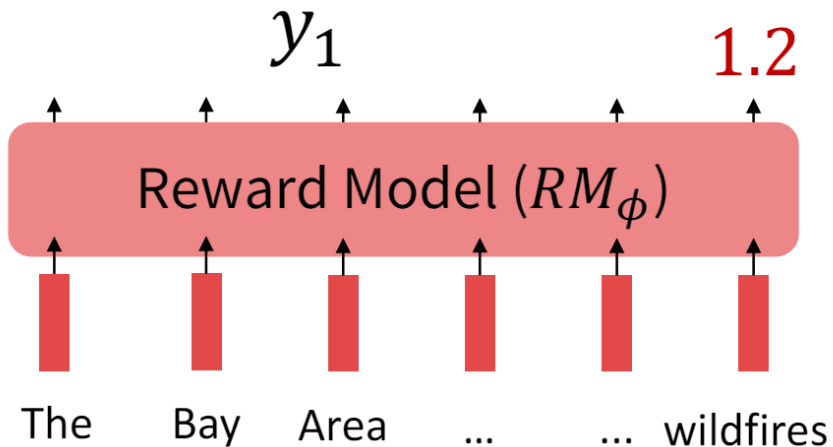
An earthquake hit
San Francisco.
There was minor
property damage,
but no injuries.

>

A 4.2 magnitude
earthquake hit
San Francisco,
resulting in
massive damage.

>

The Bay Area has
good weather but is
prone to
earthquakes and
wildfires.



y_3

Bradley-Terry [1952] paired comparison model

$$J_{RM}(\phi) = -\mathbb{E}_{(x, y^w, y^l) \sim D} [\log \sigma(RM_\phi(x, y^w) - RM_\phi(x, y^l))]$$

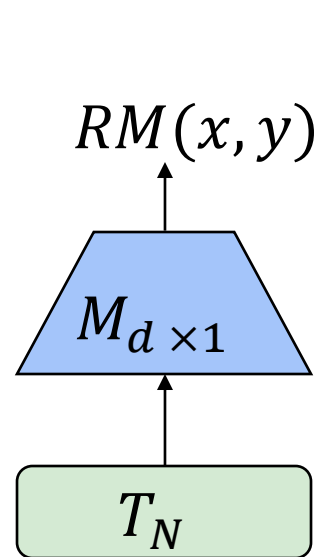
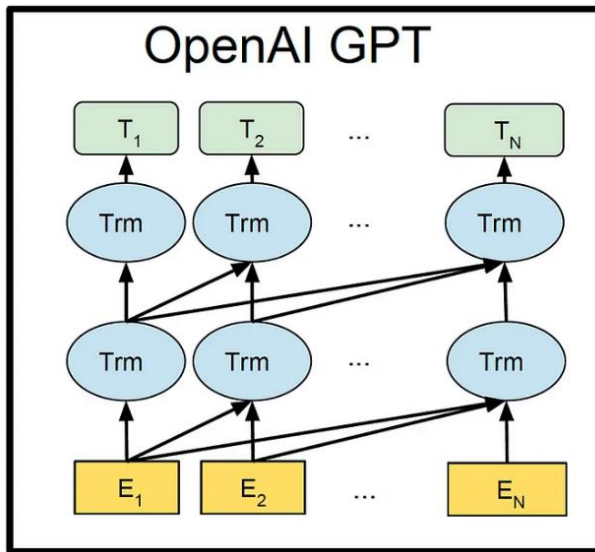
“winning”
sample

“losing”
sample

y^w should score
higher than y^l

How do we model human preferences?

- **Problem 2:** human judgments are noisy and miscalibrated!
- **Solution:** instead of asking for direct ratings, ask for **pairwise comparisons**, which can be more reliable [[Phelps et al., 2015](#); [Clark et al., 2018](#)]



Ex)

Human preference: $y_2 > y_1 > y_3$

Pairwise Labels: $y_2 > y_1, \quad y_1 > y_3, \quad y_2 > y_3$

$$\begin{aligned} Loss &= -\sum \log \sigma(RM(x, y^+) - RM(x, y^-)) \\ &= -(\log \sigma(RM(x, y_2) - RM(x, y_1))) \\ &\quad -(\log \sigma(RM(x, y_1) - RM(x, y_3))) \\ &\quad -(\log \sigma(RM(x, y_2) - RM(x, y_3))) \end{aligned}$$

RLHF: Optimizing the learned reward model

$$p(y_{1:n}|x) = \prod_{k=1}^n p(y_k|x, y_{1:k-1})$$

- We have the following:
 - A pretrained (possibly instruction-finetuned) LM $p^{PT}(y | x)$
 - A reward model $RM_{\phi}(x, y)$ that produces scalar rewards for LM outputs, trained on a dataset of human comparisons
- Now to do RLHF:
 - Copy the model $p_{\theta}^{RL}(y | x)$, with parameters θ we would like to optimize
 - We want to optimize:

$$\mathbb{E}_{\hat{y} \sim p_{\theta}^{RL}(\hat{y}|x)} [RM_{\phi}(x, \hat{y})]$$

RLHF: Optimizing the learned reward model

- We want to optimize:

$$\mathbb{E}_{\hat{y} \sim p_{\theta}^{RL}(\hat{y}|x)} [RM_{\phi}(x, \hat{y})]$$

- Do you see any problems?
 - Learned rewards are imperfect; this quantity can be imperfectly optimized
- Add a penalty for drifting too far from the initialization:

$$\mathbb{E}_{\hat{y} \sim p_{\theta}^{RL}(\hat{y}|x)} [RM_{\phi}(x, \hat{y}) - \underbrace{\beta \log \left(\frac{p_{\theta}^{RL}(\hat{y} | x)}{p^{PT}(\hat{y} | x)} \right)}]$$

Pay a price when

$$p_{\theta}^{RL}(\hat{y} | x) > p^{PT}(\hat{y} | x)$$

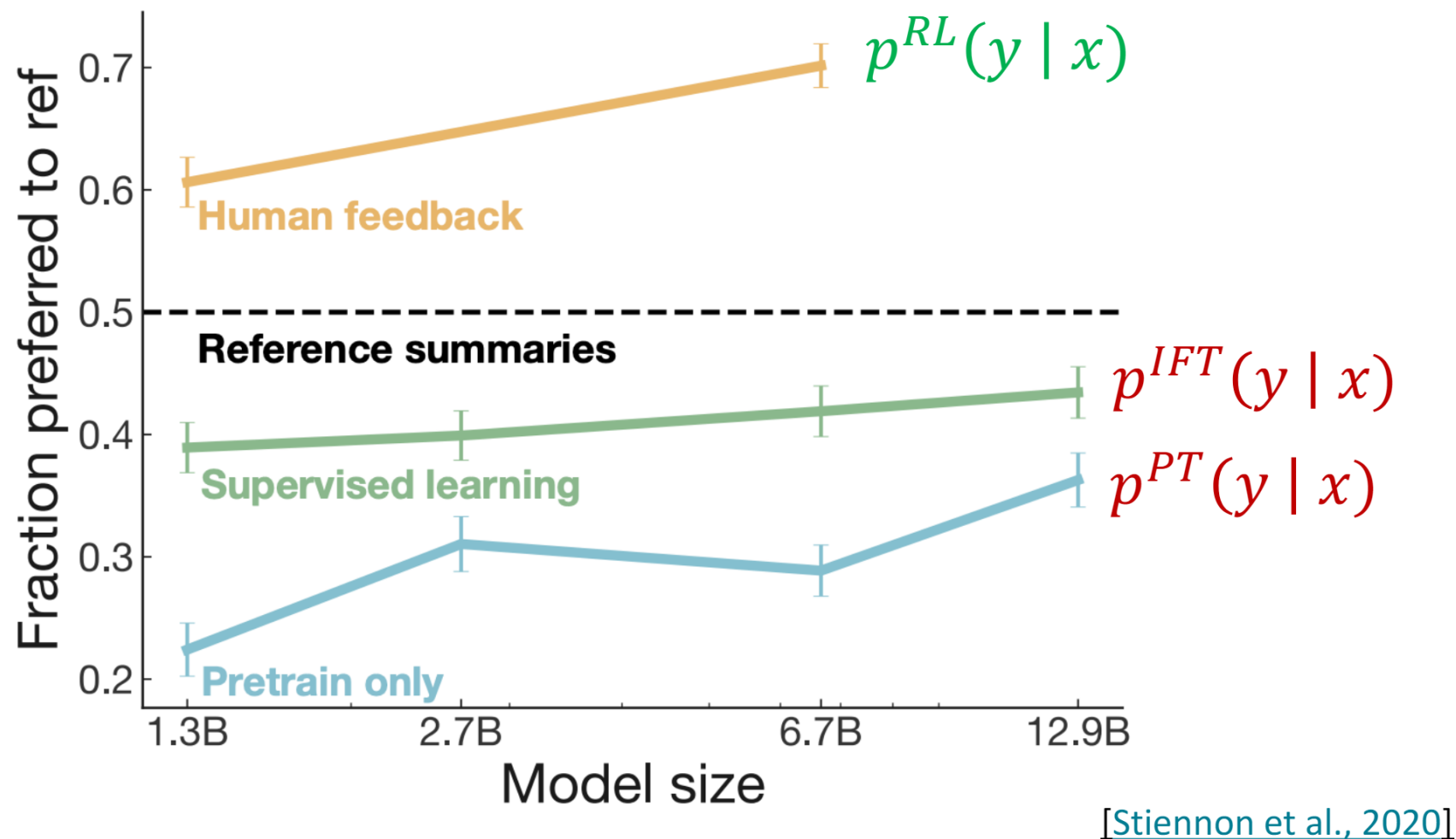
This penalty which prevents us from diverging too far from the pretrained model. In expectation, it is known as the **Kullback-Leibler (KL)** divergence between $p_{\theta}^{RL}(\hat{y} | x)$ and $p^{PT}(\hat{y} | x)$.

How to optimize? Reinforcement Learning!

- The field of **reinforcement learning (RL)** has studied these (and related) problems for many years now [[Williams, 1992](#); [Sutton and Barto, 1998](#)]
- Circa 2013: resurgence of interest in RL applied to deep learning, game-playing [[Mnih et al., 2013](#)]
- But the interest in applying RL to modern LMs is an even newer phenomenon [[Ziegler et al., 2019](#); [Stiennon et al., 2020](#); [Ouyang et al., 2022](#)]. **General Idea:**
 - Generate completions from p_{θ}^{RL} for several tasks
 - Compute reward using $RM_{\phi}(x, y)$
 - Update $p_{\theta}^{RL}(y | x)$ to increase probability of high-reward completions

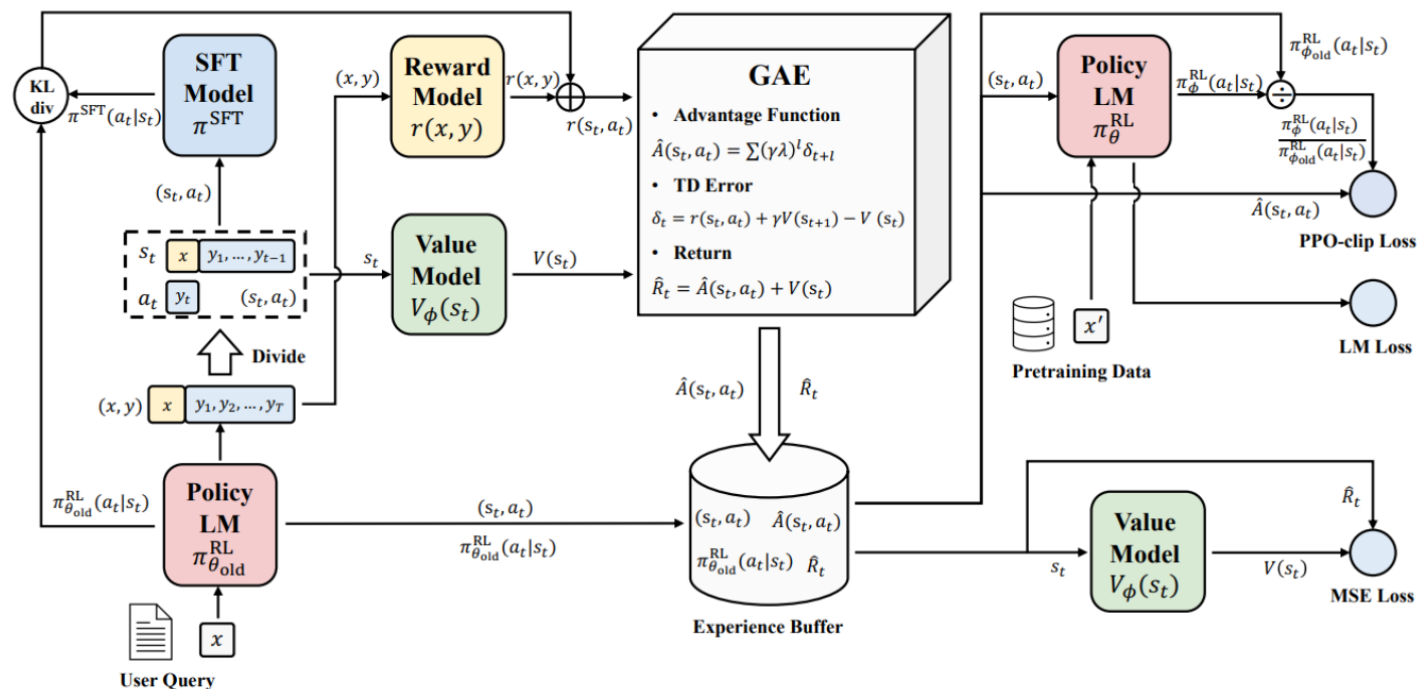


RLHF provides gains over pretraining + finetuning



RLHF can be complex

- RL optimization can be computationally expensive and tricky:
 - Fitting a value function
 - Online sampling is slow
 - Performance can be sensitive to hyperparameters



[Secrets of RLHF. Zheng et al. 2023]

Can we simplify RLHF? Towards **Direct Preference Optimization**

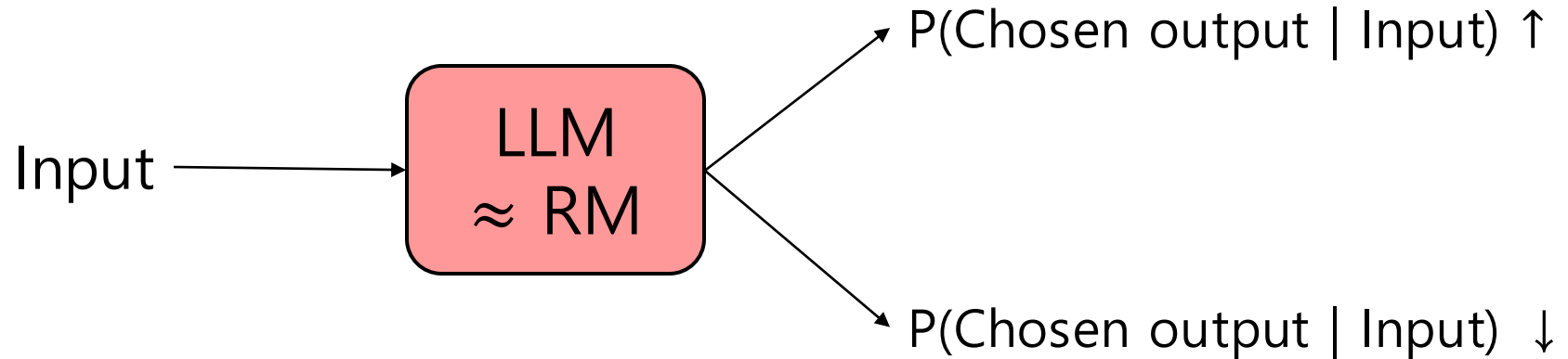
- Current pipeline is as follows:
 - Train a reward model $RM_{\phi}(x, y)$ to produce scalar rewards for LM outputs, trained on a **dataset of human comparisons**
 - Optimize pretrained (possibly instruction-finetuned) LM $p^{PT}(y | x)$ to produce the final RLHF LM $p_{\theta}^{RL}(\hat{y} | x)$
- What if there was a way to write $RM_{\phi}(x, y)$ in terms of $p_{\theta}^{RL}(\hat{y} | x)$?
 - Derive $RM_{\theta}(x, y)$ in terms of $p_{\theta}^{RL}(\hat{y} | x)$
 - Optimizing parameters θ by fitting $RM_{\theta}(x, y)$ to the preference data instead of $RM_{\phi}(x, y)$
- How is this possible? The only external information to the optimization comes from the preference labels

Can we simplify RLHF? Towards **Direct Preference Optimization**

Prepared Dataset:

(Input, Chosen output)

(Input, Rejected output)



$$P(\text{Chosen output} \mid \text{Input}) > P(\text{Rejected output} \mid \text{Input})$$

Direct Preference Optimization (DPO)

- Recall, we want to maximize the following objective:

$$\mathbb{E}_{\hat{y} \sim p_{\theta}^{RL}(\hat{y}|x)} [RM(x, \hat{y}) - \beta \log \left(\frac{p_{\theta}^{RL}(\hat{y} | x)}{p^{PT}(\hat{y} | x)} \right)]$$

- There is a closed form solution to this:

$$p^*(\hat{y} | x) = \frac{1}{Z(x)} p^{PT}(\hat{y} | x) \exp \left(\frac{1}{\beta} RM(x, \hat{y}) \right)$$

- Rearrange the terms:

$$RM(x, \hat{y}) = \beta \log \frac{p^*(\hat{y} | x)}{p^{PT}(\hat{y} | x)} + \beta \log Z(x)$$

- This holds true for arbitrary LMs

$$RM_{\theta}(x, \hat{y}) = \beta \log \frac{p_{\theta}^{RL}(\hat{y} | x)}{p^{PT}(\hat{y} | x)} + \beta \log Z(x)$$

Direct Preference Optimization (DPO)

- Recall, how we fit the reward model $RM_\phi(x, y)$:

$$J_{RM}(\phi) = -\mathbb{E}_{(x, y^w, y^l) \sim D} [\log \sigma(RM_\phi(x, y^w) - RM_\phi(x, y^l))]$$

- Notice that we only need the **difference** between the rewards for y^w and y^l . Simplify for $RM_\theta(x, y)$:

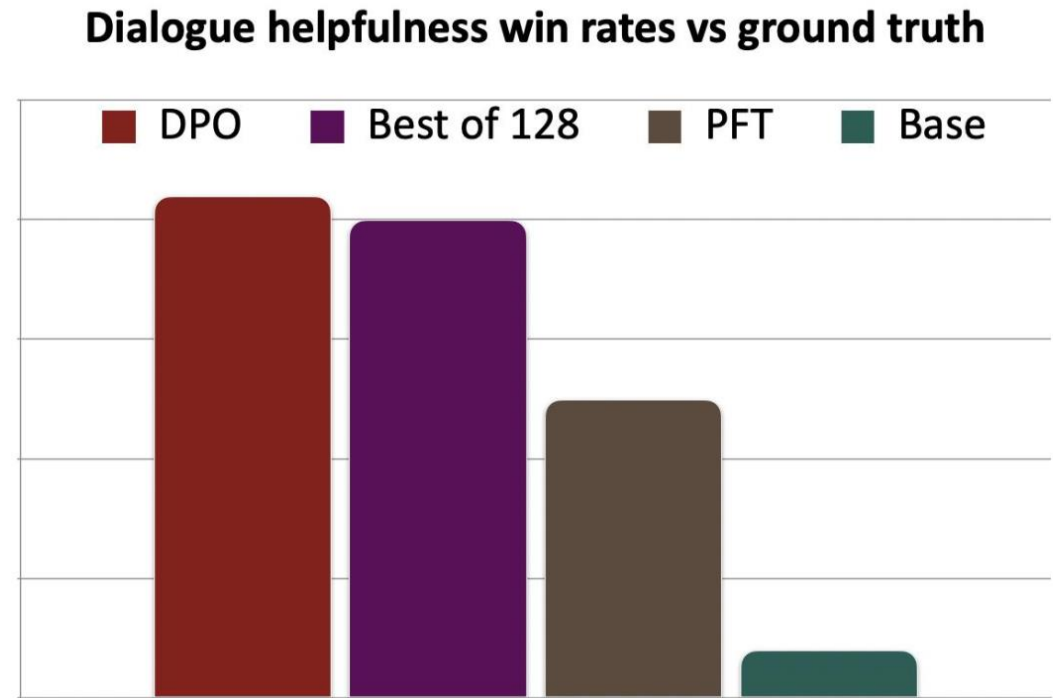
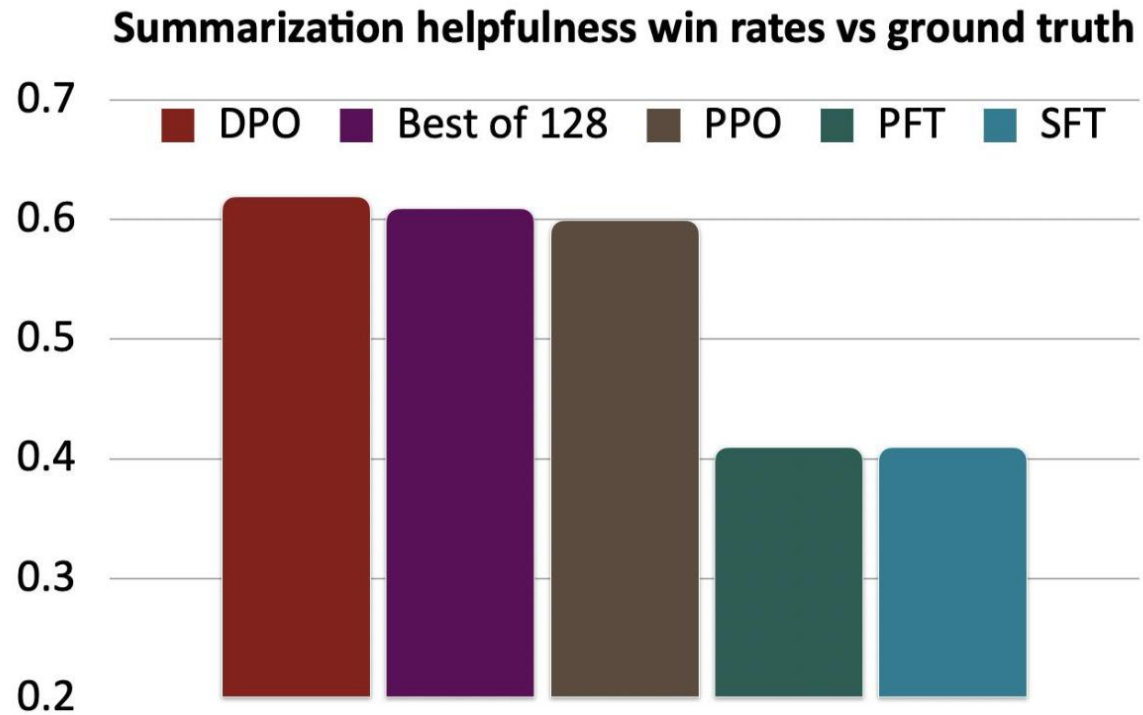
$$RM_\theta(x, y^w) - RM_\theta(x, y^l) = \beta \log \frac{p_\theta^{RL}(y^w | x)}{p^{PT}(y^w | x)} - \beta \log \frac{p_\theta^{RL}(y^l | x)}{p^{PT}(y^l | x)}$$

- The final DPO loss function is:

$$J_{DPO}(\theta) = -\mathbb{E}_{(x, y^w, y^l) \sim D} [\log \sigma(RM_\theta(x, y^w) - RM_\theta(x, y^l))]$$

We have a *simple classification loss* function that connects **preference data** to **language model parameters** directly!

Direct Preference Optimization (DPO)



Summary (RLHF and DPO)

방향

- 사람이 직접 답을 작성하거나 평가를 하는 대신, 언어 모델로 생성된 다양한 답변들의 순위를 사람이 매김
- 위 데이터를 통해 사람 선호도가 높은 답변을 생성하도록 언어모델 최적화

RLHF

- 선호도 비교 데이터에 대한 명시적 Reward 모델을 훈련하여 Reward 점수를 예측
- 예측된 Reward를 최대화하도록 언어모델 최적화 (KL 제약 추가).
- 단점: 잘 튜닝하면 효과적이지만, 계산 비용이 크고 학습 불안정.

Limitations of RL + Reward Modeling

- Human preferences are unreliable!
 - "Reward hacking" is a common problem in RL



<https://openai.com/blog/faulty-reward-functions/>

Limitations of RL + Reward Modeling

- Human preferences are unreliable!
 - "Reward hacking" is a common problem in RL
 - Chatbots are rewarded to produce responses that *seem* authoritative and helpful, *regardless of truth*
- This can result in making up facts + hallucinations

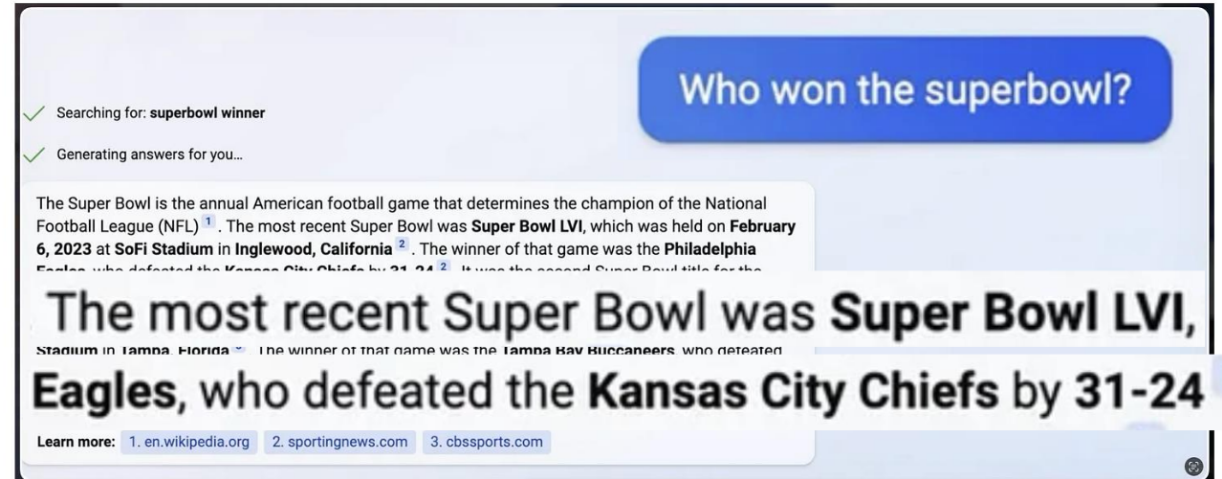
TECHNOLOGY

Google shares drop \$100 billion after its new AI chatbot makes a mistake

February 9, 2023 · 10:15 AM ET

<https://www.npr.org/2023/02/09/1155650909/google-chatbot--error-bard-shares>

Bing AI hallucinates the Super Bowl

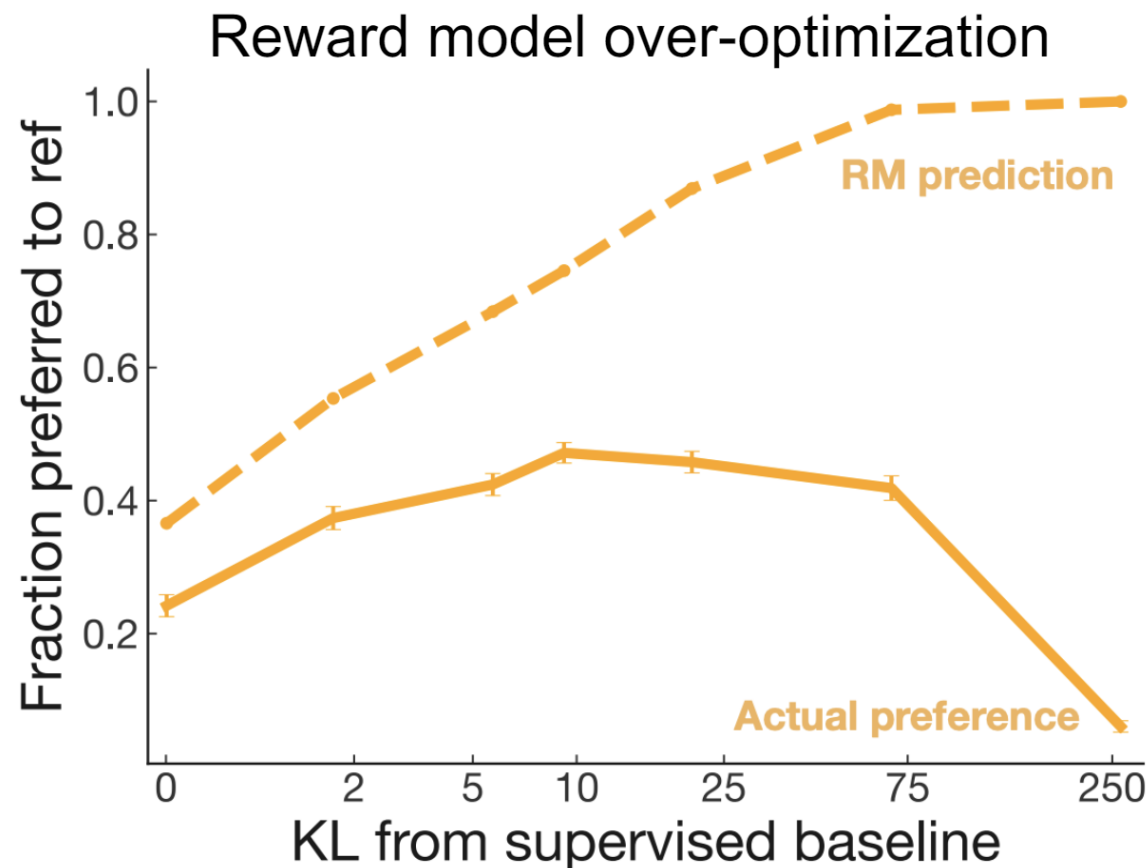


<https://news.ycombinator.com/item?id=34776508>

<https://apnews.com/article/kansas-city-chiefs-philadelphia-eagles-technology-science-82bc20f207e3e4cf81abc6a5d9e6b23a>

Limitations of RL + Reward Modeling

- Human preferences are unreliable!
 - "Reward hacking" is a common problem in RL
 - Chatbots are rewarded to produce responses that *seem* authoritative and helpful, *regardless of truth*
 - This can result in making up facts + hallucinations
- **Models** of human preferences are *even more* unreliable!



$$R(s) = RM_{\phi}(s) - \beta \log \left(\frac{p_{\theta}^{RL}(s)}{p^{PT}(s)} \right)$$

[Stiennon et al., 2020]

Open source RLHF is now mostly (not RL)

T	Model	Average	ARC	HellaSwag	MMLU	TruthfulQA	Winogrande	GSM8K
	udkai/Turdus	74.66	73.38	88.56	64.52	67.11	86.66	67.7
	fblgit/UNA-TheBeagle-7b-v1	73.87	73.04	88	63.48	69.85	82.16	66.72
	argilla/distilabeled-Marcoro14-7B-slerp	73.63	70.73	87.47	65.22	65.1	82.08	71.19
	mlabonne/NeuralMarcoro14-7B	73.57	71.42	87.59	64.84	65.64	81.22	70.74
	abideen/NexoNimbus-7B	73.5	70.82	87.86	64.69	62.43	84.85	70.36
	Neuronovo/neuronovo-7B-v0.2	73.44	73.04	88.32	65.15	71.02	80.66	62.47
	argilla/distilabeled-Marcoro14-7B-slerp-full	73.4	70.65	87.55	65.33	64.21	82	70.66
	Cultrix/MistralTrix-v1	73.39	72.27	88.33	65.24	70.73	80.98	62.77
	ryandt/MusingCaterpillar	73.33	72.53	88.34	65.26	70.93	80.66	62.24
	Neuronovo/neuronovo-7B-v0.3	73.29	72.7	88.26	65.1	71.35	80.9	61.41
	Cultrix/MistralTrixTest	73.17	72.53	88.4	65.22	70.77	81.37	60.73
	samir-fama/SamirGPT-v1	73.11	69.54	87.04	65.3	63.37	81.69	71.72
	SanjiWatsuki/Lelantos-DPO-7B	73.09	71.08	87.22	64	67.77	80.03	68.46

Handwritten notes in red:

- DPO (above udkai/Turdus)
- DPO (& UNA) (above fblgit/UNA-TheBeagle-7b-v1)
- DPO (above argilla/distilabeled-Marcoro14-7B-slerp)
- DPO (above mlabonne/NeuralMarcoro14-7B)
- Merge (of DPO models) (above abideen/NexoNimbus-7B)
- DPO (above Neuronovo/neuronovo-7B-v0.2)
- DPO (above argilla/distilabeled-Marcoro14-7B-slerp-full)
- DPO (above Cultrix/MistralTrix-v1)
- DPO (above ryandt/MusingCaterpillar)
- DPO (above Neuronovo/neuronovo-7B-v0.3)
- No info but prob DPO, given Merge (incl. DPO) (above Cultrix/MistralTrixTest)
- DPO (above samir-fama/SamirGPT-v1)
- DPO (above SanjiWatsuki/Lelantos-DPO-7B)

- Open source LLMs now almost all just use DPO (and it works well!)

Where do the labels come from?

Exclusive: OpenAI Used Kenyan Workers on Less Than \$2 Per Hour to Make ChatGPT Less Toxic
15 MINUTE READ



Millions of Workers Are Training AI Models for Pennies

From the Philippines to Colombia, low-paid workers label training data for AI models used by the likes of Amazon, Facebook, Google, and Microsoft.



Colombian Vera Fuentes with her dog. COURTESY OF OSKARINA VERO FUENTES

Behind the AI boom, an army of overseas workers in 'digital sweatshops'

By Rebecca Tan and Regina Cabello
August 26, 2023 at 2:00 a.m. EDT



- RLHF labels are often obtained from overseas, low-wage workers

From Language Models to Assistants

- Pretraining: In-context Learning
- Instruction Finetuning
- Optimizing for Human Preference (DPO/RLHF)
 - + Directly model preferences (cf. language modeling), generalize beyond labeled data.
 - RL is very tricky to get right.
 - Human preferences are fallible (imperfect); models of human preferences even more so.
- What is next?

What's next?

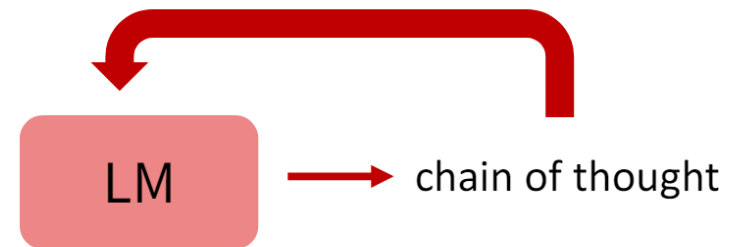
- RLHF is still a very underexplored and fast-moving area: by the next lecture (2024) these slides may look completely different!
- RLHF is (still!) data expensive. Recent work aims to alleviate such data requirements:
 - RL from **AI feedback** [[Bai et al., 2022](#)]
 - Finetuning LMs on their own outputs [[Huang et al., 2022](#); [Zelikman et al., 2022](#)]; especially code and reasoning.

LARGE LANGUAGE MODELS CAN SELF-IMPROVE

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Hongkun Yu² Jiawei Han¹

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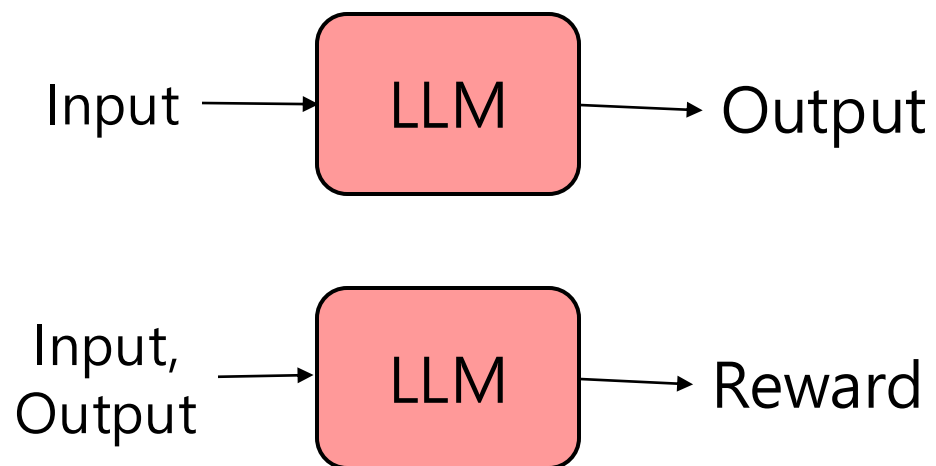
[[Huang et al., 2022](#)]



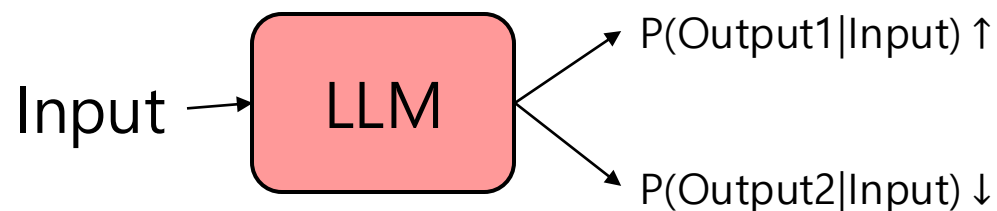
Self-Taught Reasoner (STaR)
[[Zelikman et al., 2022](#)]

What's next?

- RL from AI feedback



DPO finetuning



- Self Rewarding Prompt

Review the user's question and the corresponding response using the additive 5-point scoring system described below. Points are accumulated based on the satisfaction of each criterion:

- Add 1 point if the response is relevant and provides some information related to the user's inquiry, even if it is incomplete or contains some irrelevant content.
- Add another point if the response addresses a substantial portion of the user's question, but does not completely resolve the query or provide a direct answer.
- Award a third point if the response answers the basic elements of the user's question in a useful way, regardless of whether it seems to have been written by an AI Assistant or if it has elements typically found in blogs or search results.
- Grant a fourth point if the response is clearly written from an AI Assistant's perspective, addressing the user's question directly and comprehensively, and is well-organized and helpful, even if there is slight room for improvement in clarity, conciseness or focus.
- Bestow a fifth point for a response that is impeccably tailored to the user's question by an AI Assistant, without extraneous information, reflecting expert knowledge, and demonstrating a high-quality, engaging, and insightful answer.

User: <INSTRUCTION_HERE>

<response><RESPONSE_HERE></response>

After examining the user's instruction and the response:

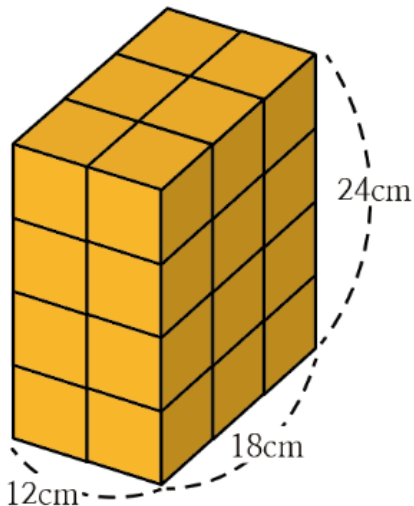
- Briefly justify your total score, up to 100 words.
- Conclude with the score using the format: "Score: <total points>"

Remember to assess from the AI Assistant perspective, utilizing web search knowledge as necessary. To evaluate the response in alignment with this additive scoring model, we'll systematically attribute points based on the outlined criteria.

What's next?

- Self-improving Capabilities

Q: 가로 길이가 12cm, 세로 길이가 18cm, 높이가 24cm 인 직육면체가 있다. 정육면체를 쌓아 위와 같은 모양의 직육면체를 만들려고 할 때, 가능한 정육면체들의 모서리의 길이 중 최대값을 구하시오.



ChatGPT

정육면체를 쌓아 위와 같은 모양의 직육면체를 만들 때, 모서리의 길이의 최대값을 구하는 것은 중요한 문제입니다.

가로, 세로, 높이가 각각 12cm, 18cm, 24cm인 직육면체를 생각해봅시다. 이를 쌓아 위와 같은 모양을 만들려면 어떻게 쌓아야 할까요? 먼저 한 개의 직육면체를 기준으로 쌓고, 그 위에 다음 직육면체를 어떻게 위치시킬지 생각해보겠습니다.

첫 번째 직육면체의 가로면이 두 번째 직육면체의 세로면과 접촉할 것입니다. 두 번째 직육면체를 위로 올릴 때, 가로면과 세로면이 서로 닿는 것이므로, 모서리의 길이는 가로, 세로, 높이의 합이 됩니다. 즉, 첫 번째 직육면체의 가로면의 길이 12cm와 두 번째 직육면체의 세로면의 길이 18cm를 더한 값이 모서리의 길이가 됩니다. 그러므로 첫 번째 두 개의 직육면체로 만들어진 모서리의 최대 길이는 $12\text{cm} + 18\text{cm} = 30\text{cm}$ 입니다.

이제 이러한 방법으로 쌓을 수 있는 직육면체들의 모서리의 길이 중 최대값을 구할 수 있습니다. 처음에는 가로면이 12cm이고, 그 다음에는 세로면이 18cm이므로, 모서리의 길이는 $12\text{cm} + 18\text{cm} = 30\text{cm}$ 입니다.

따라서 가능한 정육면체들의 모서리의 길이 중 최대값은 30cm입니다.



What's next?

• Self-improving Capabilities



You

[Question]

가로의 길이가 12cm, 세로의 길이가 18cm, 높이가 24cm 인 직육면체가 있다. 정육면체를 쌓아 위와 같은 모양의 직육면체를 만들려고 할 때, 가능한 정육면체들의 모서리의 길이 중 최대값을 구하시오.

[Answer]

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따라서 가능한 정육면체들의 모서리의 길이 중 최대값은 30cm입니다.

[Instruction]

위 답변이 어디서 틀렸는지 피드백을 생성하고, 올바른 답변을 다시 생성하시오.



ChatGPT

위 답변에서는 직육면체를 이루는 정육면체의 모서리 길이를 결정하는 과정에서 잘못된 접근 방식을 사용하였습니다. 실제 문제는 정육면체의 모서리 길이를 이용하여 주어진 직육면체의 치수(12cm, 18cm, 24cm)를 만족시키는 최대의 모서리 길이를 찾는 것입니다. 답변에서는 모서리 길이를 단순히 두 길이의 합으로 계산했으나, 이는 문제의 요구사항을 만족시키지 못합니다.

정답을 구하기 위해서는 12cm, 18cm, 24cm 각 치수가 정육면체의 모서리 길이의 배수가 되어야 합니다. 이를 통해 각 치수를 나눌 수 있는 최대의 공통된 모서리 길이를 구해야 합니다. 즉, 12cm, 18cm, 24cm의 최대공약수를 찾는 것이 정답이 됩니다.

정답 구하기

- **최대공약수(GCD):** 12, 18, 24의 최대공약수를 계산하면, 각 치수를 나눌 수 있는 최대의 모서리 길이를 구할 수 있습니다.

계산

1. **12cm와 18cm의 최대공약수**는 6입니다.
2. 이 **최대공약수 6과 24cm의 최대공약수**를 계산하면 6입니다.

따라서 가능한 정육면체들의 모서리의 길이 중 최대값은 6cm입니다. 이 모서리 길이를 사용하여 가로, 세로, 높이가 각각 12cm, 18cm, 24cm인 직육면체를 완벽하게 만들 수 있습니다.



What's next?

OpenAI is hiring developers to make ChatGPT better at coding

Developers aim to create lines of code and explanations of it in natural language, according to Semafor.

OpenAI's GPT-5: A Leap Towards Smarter AI with Enhanced Reasoning and Personalization

November 9, 2023

OpenAI Data Partnerships

Working together to create open-source and private datasets for AI training.

Thank you. Any questions?